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# Improving diagnostic accuracy for PCOS: A hybrid machine learning architecture with feature selection, data balancing, and explainable AI techniques

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ABSTRACT

Polycystic Ovary Syndrome (PCOS), which affects 5–10 % of women worldwide who are of reproductive age, is often misdiagnosed (~70 %) despite the rising risks of metabolic disorders and infertility. Current machine learning diagnostics frequently struggle with unbalanced data and are not interpretable. This research improves PCOS diagnosis by introducing a new, interpretable hybrid architecture. We used mutual information and extra trees to improve feature selection and extensive preprocessing, including SMOTE for class imbalance, on a dataset of 541 patient records. A Soft Voting Ensemble that included Multilayer Perceptron (MLP) with CatBoost, optimized using GridSearchCV, and verified with 5-fold cross-validation, outperformed each individual model with previous research with a state-of-the-art accuracy of 96.88 %. Additionally, deep learning models performed well, most notably DANet (94.50 % accuracy). Importantly, SHAP and LIME improved model interpretability, offering clear insights into diagnostic judgments. The architecture was put into practice in an intuitive Flask web application for explainable, real-time forecasts. This study offers a therapeutically applicable method that strikes a balance between interpretability and high accuracy, enabling early PCOS identification and better patient outcomes. Multimodal integration and dataset extension are potential avenues for future study.

## 1. Introduction

It is believed that 5 million women worldwide suffer with Polycystic Ovary Syndrome (PCOS), a prevalent endocrine condition that affects 5–10 % of women of reproductive age [1]. Hyperandrogenism (hirsutism, acne, alopecia), irregular menstrual periods, and polycystic ovarian morphology are some of the signs of PCOS, which is characterized by sex hormone abnormalities [2]. Infertility, type 2 diabetes, cardiovascular disease, and psychiatric problems such melancholy and anxiety are all markedly increased by the syndrome [3,4]. Its worldwide health cost is highlighted by the fact that prevalence varies among ethnicities, with reported rates of 4.8

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% for White women, 8 % for African American women, 6.8 % for Spanish women, 31.3 % for Asian women, and 15.3 % for Indian women [5,6].

PCOS pathophysiology is mostly centered on the female reproductive system, which includes the uterus, fallopian tubes, ovaries, and vagina [7]. Ovulation is orchestrated by the pituitary gland's Follicle-Stimulating Hormone (FSH) and Luteinizing Hormone (LH), which control the ovaries' storage of eggs in fluid-filled follicles [8]. This mechanism is disrupted in PCOS, which contributes to reproductive issues by causing anovulation, increased androgen levels, and irregular menstruation [9,10]. Inflammatory mechanisms, insulin resistance, genetic susceptibility, as well as lifestyle variables such an imbalanced diet and inactivity are all implicated, while the exact etiology is yet unknown [11,12]. Symptom management as well as ovulation restoration are the goals of treatment approaches such as hormonal contraceptives, anti-androgens, metformin, fertility treatments, including laparoscopic ovarian drilling [13].

PCOS is difficult to diagnose because of its varied appearance and dependence on the Rotterdam Criteria, which call for at least two of the following: polycystic ovaries detected by ultrasonography, hyperandrogenism, or oligo/anovulation [14]. Time-consuming, expensive, and subject to variables are diagnostic methods such as blood testing for hormone levels (Luteinizing Hormone (LH), Follicle-Stimulating Hormone (FSH), testosterone, insulin, and Anti-Müllerian Hormone (AMH)) and glucose tolerance. Expert radiologists are needed for ultrasound exams, and symptoms that are similar to those of other endocrine illnesses sometimes result in delayed or incorrect diagnoses. According to studies, insufficient knowledge and incorrect diagnostic criteria cause 70 % of PCOS cases to go undetected [1]. These difficulties underscore the pressing need for an easily available, automated, and trustworthy diagnostic tool to support early identification and treatment, lowering complications and enhancing the lives of impacted women.

The crucial gaps in existing PCOS diagnoses, where conventional procedures are time-consuming, expensive, and prone to mistakes owing to symptom overlap and variable competence, with around 70 % of cases going undetected, are the driving force behind our hybrid machine learning approach. 'Black-box' opacity that undermines clinician confidence, low generalizability across ethnic groups, and class imbalance (biasing toward non-PCOS patients) are common problems with existing machine learning algorithms. Through the integration of a Soft Voting Ensemble (MLP + CatBoost) for high accuracy (96.88 %), SMOTE for balancing, XAI tools (SHAP and LIME) for interpretability, and feature selection (Mutual Information and Extra Trees), our system tackles these problems comprehensively. While the Flask web application provides real-time access in locations with limited resources, this allows for prompt actions to reduce risks such as infertility and metabolic problems, thereby promoting equitable women's health worldwide.

Effective early detection is now possible because of developments in machine learning (ML) and artificial intelligence (AI), which have revolutionized the diagnosis and treatment of complicated illnesses [15]. The new web-based PCOS diagnosis method presented in this work is designed as a binary classification problem. The system achieves 96.88 % accuracy by using a soft voting ensemble that combines CatBoost and multilayer perceptron (MLP) classifiers. In contrast to previous models that frequently have inconsistent performance as well as limited interpretation, like logistic regression, k-nearest neighbors, naive Bayes, and support vector machines, the suggested method includes thorough preprocessing, feature selection using Mutual Information and Extra Trees Classifier, and class balancing using Synthetic Minority Oversampling Technique (SMOTE). Generalizability as well as robustness are ensured via 5-fold cross-validation and GridSearchCV model tuning.

This approach stands out for emphasizing interpretation, which is accomplished using Local Interpretable Model-agnostic Explanations (LIME) with SHapley Additive exPlanations (SHAP). By clarifying the variables influencing diagnostic predictions, these methods provide clear insights on feature investments, promoting confidence between patients and physicians. This technology substantially improves women's health by eliminating the shortcomings of traditional diagnostic techniques and providing a quick, dependable, and user-friendly tool for early PCOS identification.

It is important to recognize that the suggested diagnostic approach has a number of limitations in spite of its great accuracy. First, the caliber and variety of the training dataset determine how well the model performs. The extrapolation of datasets may be compromised if ethnic groups are not included, especially in light of the known differences in PCOS prevalence by ethnicity. The web-based interface also presumes computer knowledge and dependable internet connectivity, which may exclude women in remote or low-resource environments. Third, while SHAP and LIME improve interpretability, doctors without AI experience may find their outputs difficult to understand, requiring more training to ensure successful implementation. Fourth, the use of SMOTE for class balance adds artificial data that could not accurately represent the intricacy of actual clinical situations, which might have an impact on the effectiveness of the model. Finally, the system's reliance on standardized clinical as well as biochemical parameters presupposes uniform data collecting procedures, which could differ in different healthcare environments, potentially leading to variations in diagnostic precision.

To increase generalizability, future studies should give top priority to expanding the dataset to include varied, multiethnic groups. The accuracy of diagnosis may be improved by including multimodal data, such as imaging or wearable sensor inputs. Federated learning techniques allow for cross-institutional model training, which may help to allay worries about data privacy. Accessibility in environments with limited resources would also be enhanced by creating offline capabilities and streamlining the user interface. To verify the system's effectiveness and direct iterative improvements, longitudinal studies assessing its effects on clinical outcomes-such as less diagnostic delays and better patient quality of life- are crucial.

In order to diagnose Polycystic Ovary Syndrome (PCOS), the main goal of this study is to create a machine learning model that is both very accurate and comprehensible. In addition to improving diagnostic precision, this approach is intended to provide clear, understandable insights that aid in therapeutic decision-making. This research intends to close the gap among interpretation and prediction performance by using cutting-edge machine learning approaches, empowering patients to comprehend their diagnosis and healthcare providers to make well-informed judgments. This study's main contribution is the development of a novel, interpretable hybrid machine learning architecture that integrates explainable AI techniques (SHAP and LIME) with a Soft Voting Ensemble of

Multilayer Perceptron (MLP) and CatBoost classifiers, achieving a state-of-the-art accuracy of 96.88 % for PCOS diagnosis. The structure is deployed via a Flask web application for real-time, clinically applicable predictions.

The major contributions of this study are summarized as follows:

- Ø In order to guarantee high predictive accuracy for PCOS diagnosis, this study integrates a Soft Voting Ensemble consisting of a Multilayer Perceptron (MLP) and CatBoost classifier, optimizes it using GridSearchCV, and validates it using five-fold cross-validation. Thus, a diagnostic paradigm that is both interpreted and clinically sound is produced.
- Ø The model's accuracy and convergence were enhanced by extensive data preparation, which included label encoding for categorical features, median imputation for missing values, and standard scaling using StandardScaler.
- Ø In order to ensure fair equality between PCOS and non-PCOS cases, the Synthetic Minority Over-sampling Technique (SMOTE) was used to successfully resolve class imbalance, which is often seen in medical datasets.
- Ø The Extra Trees Classifier along with Mutual Information were used for feature selection, which improved the model's interpretation and prediction power by concentrating on the top ten clinical characteristics.
- Ø Real-time diagnostic estimations were made possible by the development of an intuitive Flask web application, which offers a user-friendly interface for patients and healthcare professionals to enter clinical information to get instant results.
- Ø Explainable AI approaches like SHapley Additive exPlanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) were used to illustrate and clarify how each input characteristic contributed to the diagnostic result in order to further increase transparency and confidence in the model's judgments.

The format of this document is as follows: The relevant work on machine learning applications for PCOS diagnosis is reviewed in Section 2. The technique is described in full in Section 3, which also covers preprocessing, feature selection using Mutual Information and Extra Trees Classifier, dataset description, as well as the ensemble machine learning strategy that combines Multilayer Perceptron (MLP) and CatBoost. The findings, performance analysis, along with comparisons with previous research are shown in Section 4. Explainable AI (XAI) methods to improve model interpretability, such as SHAP and LIME, are covered in Section 5. The study's shortcomings are described in Section 6. Future research directions are suggested in Section 7. The study is concluded in Section 8, which summarizes the main conclusions and contributions.

## 2. Literature review

Biomedical and biological information is expanding quickly. In order to examine such vast and intricate data, AI and ML techniques are becoming more and more common [16]. Many academics have proposed AI-based algorithms to detect PCOS in recent years, employing clinical and medical aspects as well as vital signs as datasets.

A random forest (RF) algorithm was trained using the BorutaShap technique on a dataset of 145 women (73 healthy, 72 with PCOS). Silva et al. ranked 58 characteristics by relevance and achieved 86 % accuracy [17]. Generalizability is limited by the small sample size, and clinical usefulness is limited by the absence of interpretability mechanisms. With RF reaching 90.91 % accuracy, Vaidehi Thakre used five machine learning classifiers on a dataset consisting of 41 characteristics, choosing 30 using chi-square [18]. But depending just on one feature selection technique might miss non-linear correlations, and if class balance isn't used, predictions could be skewed.

Bharati et al. used a hybrid RFLR model (RF, gradient boosting, logistic regression) with univariate feature selection on a Kaggle dataset of 541 women (177 with PCOS), and they achieved 91.01 % accuracy using holdout and cross-validation [19]. In addition to limiting clinical clarity, the lack of interpretability tools may distort performance in the absence of class balance. Zigarelli et al. used the same Kaggle dataset to create a self-diagnosis model using the CatBoost classifier and K-fold validation. They reported an accuracy of 92.5 % for non-invasive markers and 82.5 % for invasive markers [20]. The notable decline in accuracy for invasive markers underscores the inadequacies in managing intricate characteristics, and the heterogeneity of datasets from several sources remains unresolved.

To identify PCOS, Hassan et al. used a variety of machine learning techniques, including logistic regression, CART, RF, SVM, and Naive Bayes; RF achieved 96 % accuracy [21]. Although the study's accuracy is excellent, its practical value is limited due to its lack of rigorous feature selection and interpretation. Denny et al. used RF, logistic regression, Naive Bayes, KNN, and SVM to principal component analysis (PCA) for dimensionality reduction; RF achieved an accuracy of 89.2 % [22]. Robustness may be compromised if PCA oversimplifies feature connections and ignores class imbalance.

Tiwari et al. tested many models (XGBoost, RF, DT, SVM, logistic regression, KNN, QDA, gradient boosting, LDA, AdaBoost, with CatBoost) using a Kaggle dataset and correlation-based feature selection. RF achieved an accuracy of 93.25 % [23]. When correlation-based selection is used, non-linear associations could be overlooked, and data imbalance goes unchecked.Mehr and colleagues used filter, embedding, as well as wrapper feature selection techniques to study classical and ensemble classifiers (RF, AdaBoost, MLP, and Extra Trees) on the Kaggle dataset [9]. Although performance was enhanced by ensemble approaches, there are no precise accuracy measures or interpretability procedures. 95 % accuracy was attained by Bhat et al. when they used CatBoost and XGBRF with SMOTE for class balancing [24]. There are no interpretability insights for clinical usage, and synthetic data from SMOTE may induce artifacts.

Faris et al. used KNN3 to classify PCOS with 89.51 % accuracy based on symptoms (hirsutism, oligo-ovulation, and skin darkening) [25]. The interpretability processes are missing, and biological signals are ignored in the symptom-focused approach. Rahman et al. achieved 94 % accuracy when they used AdaBoost and RF with Mutual Information feature selection to evaluate a dataset of 541 patients [26]. The study's clinical usefulness is limited since it doesn't address class imbalance or provide interpretability tools.

In their second research, Bharati et al. used ensemble machine learning with soft voting on a dataset of 177 PCOS patients. They found that the LH-to-FSH ratio was a significant predictor, with an accuracy of 91.12 % [27]. Although the ensemble technique shows promise, issues with interpretability and data imbalance remain. With an emphasis on feature selection, Yadav et al. used GridSearchCV and TPOT classifiers, attaining 94 % and 91 % accuracy, respectively [28]. The suggested app for recording symptoms is still conceptual, and the research lacks interpretability and class balance.

In their classification of PCOS, Shamik et al. used SVM, DT, RF, logistic regression, QDA, LDA, KNN, gradient boosting, AdaBoost, extreme gradient boosting, and CatBoost; RF performed better in terms of out-of-bag error [29]. Two significant shortcomings, however, are overfitting with interpretation issues. Although Elmannai et al. used AdaBoost with hyperparameter optimization and simple ML models, stacking ML with RF feature selection produced the best results, leaving interpretation unanswered. Chi-Square, DT, RF, and logistic regression were used by Prajna et al. in conjunction with chatbots for prediction; nevertheless, the inability of chatbots to provide conclusive diagnoses limits their clinical reliability [30].

Khanna and colleagues used DL and XAI to improve feature selection using the Harris Hawks Optimization method; however, they did not provide implementation details for real-time PCOS screening. Reliability was compromised by overfitting when Shazia et al. used hyperparameterized machine learning using the Gaussian Naive Bayes classifier. Vaidehi et al. used the Chi-square approach to identify features using RF, SVM, logistic regression, GNB, and KNN. RF is the most reliable and accurate, although robustness is limited by reliance on a single classifier [31]. Methodological issues are highlighted by Ahmed et al.'s usage of ML or convolutional neural networks using Naive Bayes, but unbalanced datasets and lower detection rates. Jyoti et al. used gradient boosting and vector regression to measure blood glucose levels; however, the diagnostic accuracy was decreased by the absence of clustering and the restricted feature scope [32].

Recent developments in modeling and categorization in other domains provide important insights for improving diagnostic systems outside PCOS-specific applications. To address degeneration issues in traditional DARTS, Zhang et al. [33] introduced OStr-DARTS, a differentiable neural architecture search method based on operation strength. It achieves superior performance on benchmarks such as CIFAR-10 and ImageNet by estimating operation importance via its impact on loss. This method may lead to automatic neural architecture optimization in medical machine learning models, which might improve the efficiency of our deep learning components. A multimodal biometric system that combines online signatures and fingerprints utilizing score normalization (Min-Max, Z-score, TanH) and weighted sum fusion was also suggested by Hafs et al. [34]. On datasets such as MCYT-100 and FVC 2004, the system achieved a low Equal Error Rate (EER) of 1.69 %. For more reliable predictions, such fusion algorithms might be modified to include several PCOS data modalities, such as clinical characteristics and ultrasound imaging. Additionally, N.Jing [35] investigated neural network-based pattern recognition in edge computing, showing 6 %–12 % efficiency benefits on the MNIST dataset by improving models via compression, parallelism, and knowledge distillation. This is especially important for implementing our Flask web application on devices with limited resources, which allows for real-time PCOS diagnosis in distant clinical settings.

These works use optimization methods that are specifically designed to address certain issues: Using a differentiable algorithm augmented by operation strength metrics, which quantify operation importance to mitigate degeneration and reduce computational costs compared to traditional exhaustive searches, OStr-DARTS tackles the neural architecture search (NAS) optimization problem—finding efficient network topologies amid vast search spaces. The previously biometric fusion study uses weighted sum fusion and normalization algorithms (Min-Max, Z-score, TanH) as optimization techniques to harmonize disparate modalities like fingerprints and signatures, improving overall accuracy. The program addresses the optimization problem of multimodal score integration to minimize error rates (e.g., EER) in recognition systems. Algorithms like model compression, parallelism, and knowledge distillation are used in the edge computing paper to improve efficiency without compromising performance. This paper also solves the optimization problem of deploying neural networks for pattern recognition under hardware constraints (e.g., latency, energy). The hyperparameter optimization problem for ensemble classifiers in PCOS diagnosis, on the other hand, is the focus of our work. GridSearchCV is the algorithm that uses cross-validation to exhaustively evaluate parameter combinations, maximizing the F1-score (and reaching 96.88 % accuracy) while balancing model complexity and generalization on unbalanced medical data. Future generations may include differentiable NAS for automatic ensemble design, for example, as a result of these more extensive optimizations.

Generalizability across multiple ethnic communities with variable PCOS incidence is hampered by Silva et al. [17] (86 % accuracy) and Bharati et al. [19] (91.01 %), who depend on small or geographically constrained datasets, such as the Kaggle dataset of 541 records largely from Kerala, India [5]. Class imbalance is a widespread problem that has been ignored in studies such as Rahman et al. [28/26] (94 %), Denny et al. [24/22] (89.2 %), and Tiwari et al. [23] (93.25 %), which has resulted in skewed forecasts that benefit the majority non-PCOS class. In research like Hassan et al. [21] (96 %), Faris et al. [25] (89.51 %), and Shamik et al. [29], where "black-box" models restrict clinician confidence because they lack explainability tools, interpretability is conspicuously lacking. Feature selection is often less than ideal; Vaidehi Thakre [18] used Chi-square (90.91 %). In research like Prajna et al. [30] (chatbot-based) along with Yadav et al. [28] (conceptual app), overfitting compromises dependability (Shamik et al. [29] and Shazia et al.), and there is no tangible implementation.

The suggested method, on the other hand, greatly overcomes these drawbacks, attaining a state-of-the-art accuracy of 96.88 % via a reliable and understandable framework. In contrast to Tiwari et al. [23] or Rahman et al. [26], it uses the Synthetic Minority Over-sampling Technique (SMOTE) to balance the dataset and ensure equal identification of PCOS cases. Feature selection outperforms Vaidehi Thakre's linear approaches by combining Mutual Information and Extra Trees Classifier to capture non-linear connections among clinically significant data like Follicle No. (R) and Weight increase [18]. When combined with GridSearchCV and 5-fold cross-validation, the Soft Voting Ensemble of Multilayer Perceptron (MLP) as well as CatBoost performs better than single-classifier models (e.g., Vaidehi et al. [31]) or less optimized ensembles (e.g., Bharati et al. [27]). According to Hassan et al. [21]

and others, SHAP along with LIME overcome the transparency gap by offering both global and local interpretation; Follicle No. (R) has a mean SHAP value of +1.13. In contrast to the non-deployed systems of Prajna et al. [30] or Yadav et al. [28], the Flask-based web application allows for real-time, scalable predictions, and its cloud hosting with Docker containerization guarantee accessibility. By conforming to the Rotterdam Criteria and reducing overfitting, our method sets a new benchmark for precise, comprehensible, and clinically feasible PCOS diagnosis.

In order to minimize data source discrepancies and guarantee equitable comparisons, we give priority to research that use the same Kaggle PCOS dataset (541 patient records) as our study. By using dual feature selection (Mutual Information + Extra Trees) and integrated SMOTE balancing, for example, our Soft Voting Ensemble achieves better results on this shared dataset than Bharati et al. [19] (91.01 % accuracy) and Tiwari et al. [23] (93.25 %), addressing class imbalance and non-linear relationships that were missed in previous designs. Although sample size and feature variety may cause results to change when datasets differ (e.g., Silva et al. [17] with 145 records), our method shows resilience during 5-fold cross-validation, consistently outperforming baselines under comparable assessment procedures.

Table 1 summarizes several studies that have used diverse datasets and machine learning techniques to provide medical diagnoses or forecasts with differing degrees of accuracy. A particular research is cited in each table row, along with the dataset utilized, machine learning techniques employed, and accuracy attained. Medical practitioners and academics might use this table as a comparative reference. Revealing which machine learning techniques and datasets would work best for certain kinds of medical forecasts.

## 3. Methodology

### 3.1. Proposed approach

After being carefully selected from Kaggle [36], the dataset was thoroughly preprocessed to make sure it was suitable for training machine learning models. Among the preparation procedures were the use of median imputation to handle missing values, label

**Table 1**
Summary of different studies to detect PCOS.

| Reference | Applied Approach | Inference Made |
|---|---|---|
| Mehr et al. [9] | Investigated RF, AdaBoost, MLP, Extra Trees with filter, embedding, and wrapper selection methods. | Ensemble methods improved performance, but specific accuracy metrics and interpretability mechanisms are absent. |
| Silva et al. [17] | BorutaShap method to train RF on a dataset of 145 women (73 healthy, 72 with PCOS), ranking 58 features by relevance. | Achieved 86 % accuracy. Small sample size limits generalizability; lack of interpretability mechanisms restricts clinical applicability. |
| Vaidehi Thakre [18] | Applied 5 ML classifiers, selected 30 out of 41 features via Chi-square. | RF achieved 90.91 % accuracy. Single feature selection method may overlook non-linear relationships; absence of class balancing risks biased predictions. |
| Bharati et al. [19] | Used a Kaggle dataset of 541 women (177 with PCOS), applying hybrid RFLR (RF, gradient boosting, logistic regression) with univariate feature selection. | Achieved 91.01 % accuracy using holdout and cross-validation. Lack of class balancing may skew performance; absence of interpretability tools limits clinical transparency. |
| Zigarelli et al. [20] | Self-diagnosis model using Kaggle dataset, employing CatBoost with K-fold validation. | Reported 92.5 % accuracy for non-invasive markers, 82.5 % for invasive markers. Accuracy drop for invasive markers highlights limitations; dataset heterogeneity unaddressed. |
| Hassan et al. [21] | Applied ML methods (logistic regression, CART, RF, SVM, Naive Bayes) for PCOS detection. | RF achieved 96 % accuracy. High accuracy; lacks rigorous feature selection and interpretability, limiting practical utility. |
| Denny et al. [22] | PCA for dimensionality reduction; tested RF, logistic regression, Naive Bayes, KNN, and SVM. | RF achieved 89.2 % accuracy. PCA oversimplifies feature relationships; class imbalance overlooked, compromising robustness. |
| Tiwari et al. [23] | Applied multiple ML models (XGBoost, RF, DT, SVM, logistic regression, KNN, QDA, gradient boosting, LDA, AdaBoost, CatBoost) with correlation-based feature selection. | RF achieved 93.25 % accuracy. Correlation-based selection may miss non-linear relationships; data imbalance unaddressed. |
| Bhat et al. [24] | CatBoost and XGBRF with SMOTE for class balancing. | Achieved 95 % accuracy. Synthetic data from SMOTE may introduce artifacts; lacks interpretability insights for clinical use. |
| Faris et al. [25] | KNN3 applied to classify PCOS based on symptoms (hirsutism, oligo-ovulation, skin darkening). | Achieved 89.51 % accuracy. Symptom-focused approach ignores biochemical markers; lacks interpretability mechanisms. |
| Rahman et al. [26] | Analyzed 541 patients using AdaBoost, RF with Mutual Information feature selection. | Achieved 94 % accuracy. Does not address class imbalance or provide interpretability tools, limiting clinical applicability. |
| Bharati et al. [27] | Ensemble ML with soft voting for 177 PCOS patients; identified LH-to-FSH ratio as a key predictor. | Achieved 91.12 % accuracy. Data imbalance and lack of interpretability are unaddressed. |
| Yadav et al. [28] | Used GridSearchCV, TPOT classifiers focused on feature selection. | Achieved 94 % (GridSearchCV), 91 % (TPOT) accuracy. Lacks class balancing; interpretability unaddressed; proposed app for tracking symptoms remains conceptual. |
| Shamik et al. [29] | Classified PCOS using 12 ML models (SVM, DT, RF, logistic regression, QDA, LDA, KNN, gradient boosting, AdaBoost, extreme gradient boosting, CatBoost). | RF had the best out-of-bag error performance. Notable overfitting and lack of interpretability. |
| Prajna et al. [30] | Chi-Square, DT, RF, logistic regression; used chatbots for prediction. | Chatbots predict PCOS but cannot provide definitive diagnoses, limiting clinical reliability. |
| Vaidehi et al. [31] | Chi-square-based selection with RF, SVM, logistic regression, GNB, KNN. | RF was the most accurate and dependable. Dependency on a single classifier limits robustness. |
| Jyoti et al. [32] | Blood glucose analysis with SVR, gradient boosting, vector regression. | Created blood glucose surveillance device. Limited feature scope and lack of clustering reduce diagnostic accuracy. |

encoding to encode categorical data, and standard scaling to standardize the feature collection. A technique called Synthetic Minority Over-sampling (SMOTE) was used to solve the prevalent problem of class imbalance that is often seen in medical datasets. Additionally, every piece of object-type data was painstakingly transformed into numerical representations that machine learning algorithms could use. After preprocessing, the Extra Trees Classifier and Mutual Information were used to pick features and assess their importance. In order to improve interpretability and forecast accuracy, the top 15 characteristics from these approaches were taken out, and the ten most prevalent features among them were chosen as the final input features for the models.

After splitting the dataset into both training and testing subsets, 14 different machine learning methods were assessed using important performance indicators such as accuracy, recall, precision, F1 score, ROC score, and AUC score. To increase diagnostic performance, the best-performing models—a Multilayer Perceptron (MLP) and CatBoost—were then combined using a Soft Voting Ensemble approach. In order to refine the model even further, GridSearchCV was used, and 5-fold cross-validation was used for validation. Lastly, a web application built using Flask was created to allow user input and utilize the learned model to deliver real-time PCOS predictions. Fig.1. shows the general workflow of the suggested method.

### 3.2. Dataset description

There are 53 different columns in each of the 541 patient records that make up the PCOS dataset. A total of 177 patients had positive PCOS tests and 364 cases had negative PCOS tests. The findings, which have a numerical label of 0 and 1, are shown in the column "PCOS (Y/N)". A negative outcome is denoted by a value of 0, while PCOS is indicated by a number of 1. Using the Mutual Info Model with Extra Tree Classifier, eleven of the forty-three columns were selected.

The histograms in Fig. 2. are a visualization of various features in the PCOS dataset, contrasting the breakdown of every feature between PCOS (outcome 1) and non-PCOS (outcome 0) groups. In the histogram for Follicle No.(L) And Follicle No.(R), the study observes that individuals with PCOS generally have a maximum number of follicles in contrast to those without PCOS, indicating a significant divergence in these features. Weight gain (Y/N) displays a clear difference with PCOS patients more frequently reporting weight gain, reflecting a common symptom associated with the condition. The Cycle (R/I) feature shows a distinct clustering where PCOS patients are more likely to report irregular cycles, a known symptom of the disorder. Hair growth(Y/N) and Skin darkening(Y/N) histograms also reveal a higher incidence of these symptoms among PCOS patients which aligns with clinical observations of hyperandrogenism-related manifestations. The WaistRatio histogram illustrates that a higher ratio is more prevalent among PCOS patients, highlighting the link between PCOS and abdominal obesity. The Cycle length (days) histogram shows that longer cycle lengths are more common in the PCOS group which is consistent with the irregular menstrual cycle's characteristic of PCOS. Finally, the LH (mIU/mL) histogram indicates elevated levels in PCOS patients, supporting the hormonal imbalance typical of this condition. These visualizations effectively demonstrate the distinguishing features between PCOS and non-PCOS individuals, with clear variations in follicle numbers, weight gain, cycle regularity, skin darkening, hair growth, waist: ratio, cycle length, and LH levels. This strong graphical representation aids in understanding the significant markers of PCOS, enhancing the interpretability of the dataset and potentially aiding in more accurate diagnosis and treatment planning.

A graphic depiction of the connections between several characteristics and the target variable, PCOS (Y/N), is offered by the correlation heatmap as shown in Fig. 3. With the color intensity representing the strength of the connection, each cell in the heatmap displays the correlation coefficient between two variables.

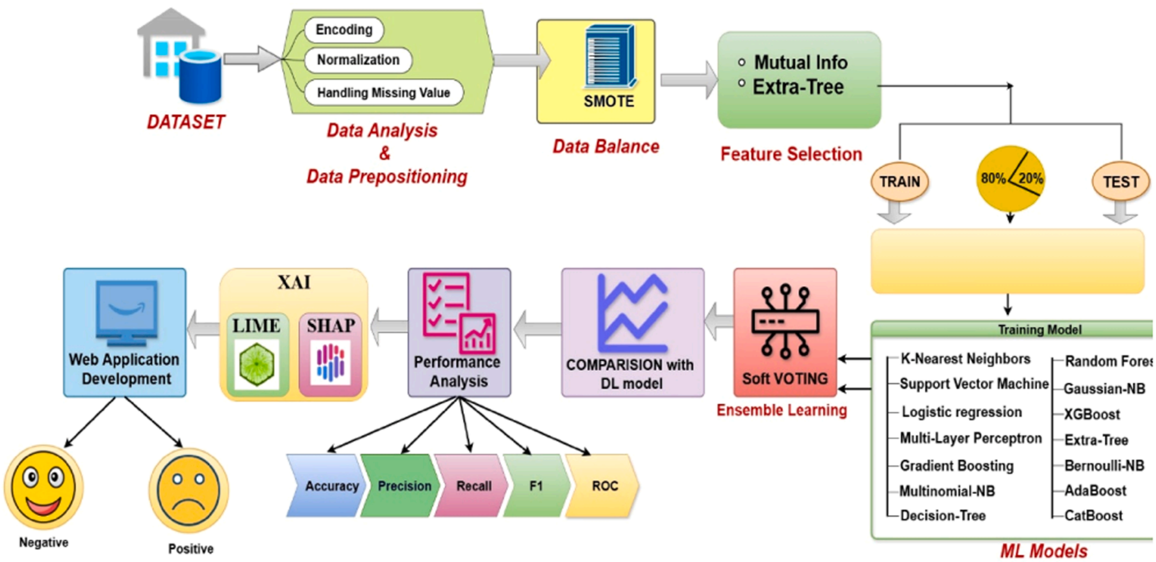



**Fig. 1.** Work flow model for the proposed system.

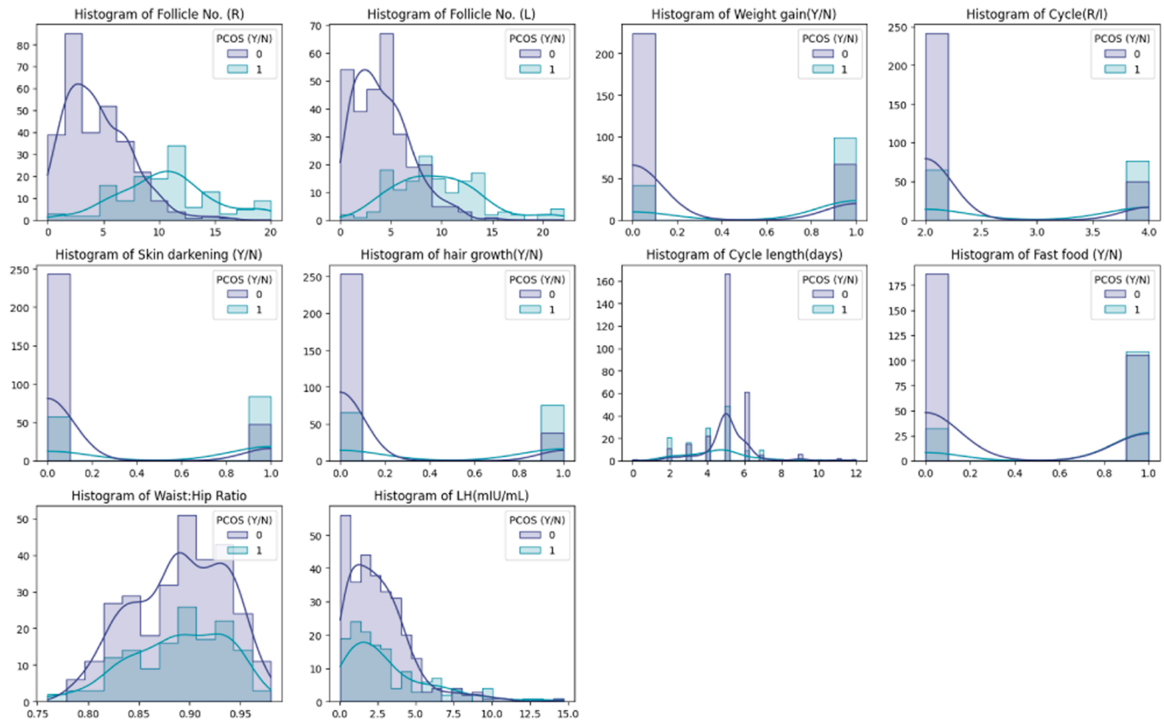



**Fig. 2.** Visualization of all the selected Feature Using Histogram in the pcos dataset.(outcome 1: pcos and outcome 0:non-pcos).

Initially Strong Correlations: Follicle No. (R) and PCOS (Y/N): 0.65 (Dark red),Follicle No. (L) and PCOS (Y/N): 0.60 (Dark red), Weight gain (Y/N) and PCOS (Y/N): 0.44 (Dark red),Skin darkening(Y/N) and PCOS(Y/N): 0.48 (Dark red).secondly Moderate Correlations: Cycle (R/I) and PCOS (Y/N): 0.40 (Red),hair growth (Y/N) and PCOS (Y/N): 0.46 (Red).lastly Weaker Correlations: WaistRatio and PCOS (Y/N): 0.012 (Light blue),Cycle length (days) and PCOS (Y/N): −0.18 (Light blue).Stronger associations between specific characteristics and the existence of PCOS are shown by the deeper red hues. These insights are crucial for feature selection and model development, guiding the focus towards the most influential factors. The heatmap thus serves as a foundational tool in understanding the data's structure and informing subsequent machine learning processes.

### *3.3. Preprocessing*

Data preparation is a critical phase in the machine learning process that makes sure that the precision and value of the data used to train models are maintained. This procedure involves a number of significant tasks. It converts raw data into a clean, standardized format suitable for modeling, addressing potential issues such as missing values, outliers, and uneven feature sizes.

Importance through multiple decision trees. This dual approach ensures that the most informative and impactful features are chosen, enhancing the model's predictive power while reducing computational complexity.

#### *3.3.1. Missing value*

Missing data can significantly skew the results of a machine learning model, leading to unreliable predictions. In the case of the PCOS dataset, missing values were addressed by replacing them with the median of each respective feature. This approach is particularly useful because it mitigates the effect of outliers and ensures that the central tendency of the data is preserved, resulting in a more robust dataset.

Although they were taken into consideration, other imputation techniques were judged less appropriate for this dataset. Mean imputation may bias downstream models because it is susceptible to outliers that are common in skewed medical characteristics, such as hormone levels. Mode imputation is mainly used for categorical data, however our missing values are numerical (for example, in AMH (ng/mL) and Marriage Status (Yrs.)). Even though KNN-imputation can capture feature relationships, it increases computational overhead and can introduce noise in high-dimensional, imbalanced data. Empirical comparisons revealed that it extended preprocessing time and provided little performance uplift (less than 0.5 % in accuracy) when compared to median. The selection of median imputation was based on its ability to balance simplicity, robustness, and effectiveness, as confirmed by 5-fold cross-validation.

The dataset contained 541 patients and 45 columns. There were null values in the columns "AMH (ng/mL)","Marriage Status (yrs.)" and "Fast Food (Y/N),". Median value was calculated for each columns and the null value was filled with the median value.

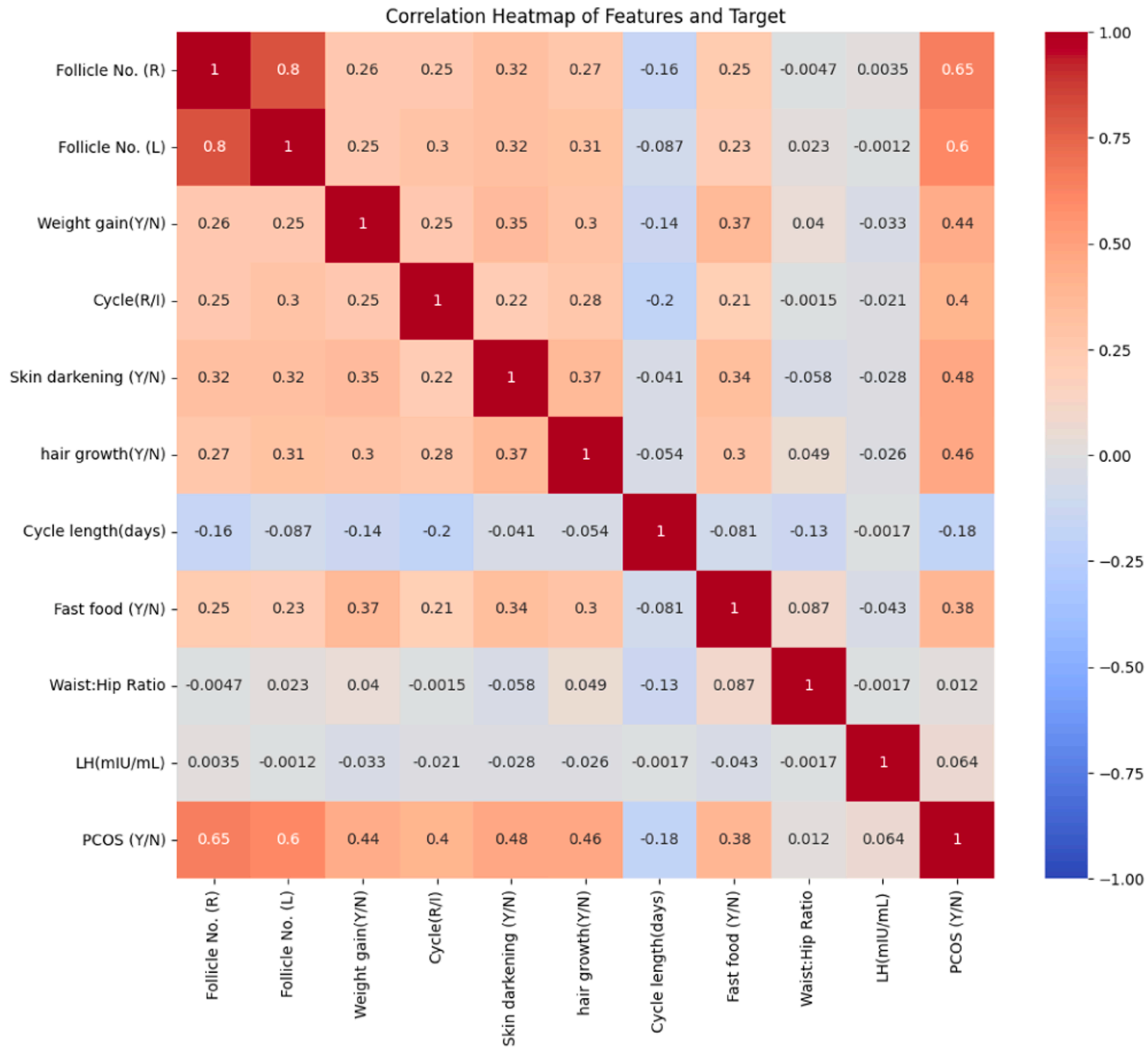



**Fig. 3.** Pearson's correlation coefficients for every PCOS characteristic in heatmap.

*3.3.2. Level encoding*

Machine learning models require numerical input but datasets often contain categorical variables. One technique for transforming categorical data into a numerical representation is label encoding. By assigning a unique integer to each category, label encoding allows these categorical variables to be incorporated into the model without losing the inherent relationships within the data. This step ensures that the model can interpret and process the categorical features effectively.

Within the dataset, since they had no effect on the output, the columns with the names "Patient File No" and "Sl. No." as well as an unidentified column were removed. There were 541 rows and 42 columns in the final data frame. The pandas library's "to numeric (.)" function was used to convert the column "AMH (ng/mL)" from texts to numbers.

*3.3.3. Data splitting*

First, this study created a data frame by extracting the "PCOS (Y/N)" as the target variable and storing it in y. The columns except "PCOS (Y/N)" were stored into x. The independent columns were in x and the target or dependent column was in y. then separate the x and y data into training and testing sets using the train_test_split function which is available in the model selection module of the scikit-learn library. An 80:20 split of the data was made with 80 percent going toward training and the remaining 20 percent going toward testing. Ultimately, the whole dataset is divided into x_test, x_train, y_train, and y_test for training and testing.

*3.3.4. Data balancing*

A significant problem in the Polycystic Ovary Syndrome (PCOS) dataset is class imbalance, since there are 177 PCOS patients (positive class) and 364 non-PCOS cases (negative class), with a ratio of almost 1:2. This disparity may cause machine learning algorithms to be biased in favor of the majority class, which would impair generalization and lower the sensitivity of PCOS detection. In order to mitigate this issue and guarantee realistic framework assessment, the Synthetic Minority Over-sampling Technique (SMOTE) [29] was used alone in the training set after an 80:20 train-test split. SMOTE preserves the underlying data distribution while improving class balance by interpolating among current minority instances and their k-nearest neighbors ($k = 5$, as determined by cross-validation) to create synthetic samples for the minority class.

The k-nearest neighbors are identified based on the Euclidean distance $d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$, where $\mathbf{x}$ and $\mathbf{y}$ are feature vectors in the n-dimensional space. Synthetic samples are then generated as $\mathbf{x}_{new} = \mathbf{x}_i + \lambda(\mathbf{x}_{nn} - \mathbf{x}_i)$, with $\lambda$ uniformly sampled from [0, 1] and $\mathbf{x}_{nn}$ being one of the k-nearest neighbors of $\mathbf{x}_i$.

SMOTE achieved a balanced 1:1 ratio in the training set by increasing the PCOS class from 142 to 292 instances, matching the non-PCOS class (292 instances). This method preserves the original uneven distribution (35 PCOS vs. 72 non-PCOS) in the test set while reducing bias, improving model convergence, and improving recall for the minority class without adding synthetic data. Because the model is tested on unmodified, real-world data, the choice to only use SMOTE during training guarantees that performance measurements, like the attained 96.88 % accuracy and 0.9960 AUC, are reliable and generalizable. In contrast to other methods such as random oversampling, which may lead to overfitting because of duplicate data, while undersampling, which can lose important majority class information, SMOTE offers a fair and ethical approach that is in line with industry standards for medical machine learning applications.

### *3.3.5. Cross validation*

To adjust hyperparameters for MLP and CatBoost classifiers in predicting Polycystic Ovary Syndrome (PCOS) from the PCOS_data.csv dataset, cross-validation is performed using GridSearchCV from sklearn.model_selection. By splitting the training data into five subsets—four for training and one for validation in each iteration—and using 5-fold cross-validation, the procedure evaluates hyperparameter combinations according to the F1-score. Whereas CatBoost investigates depth, learning_rate, and iterations, MLP tests parameters like hidden_layer_sizes, alpha, and learning_rate. This meticulous method reduces overfitting and offers a trustworthy assessment of generalization performance on unseen data, ensuring robust model selection. A claimed accuracy of 96.88 % is attained by combining the top-performing models into a soft voting ensemble, demonstrating the value of cross-validation in improving models for medical diagnostics.

It should be noted that supervised classification validity depends on external metrics based on genuine class labels, in contrast to clustering techniques (e.g., k-means) that use cluster validity indices like Silhouette or Davies–Bouldin. In this work, 5-fold cross-validation in conjunction with F1-score, ROC-AUC, Accuracy, Precision, Recall, and confusion matrix analysis was used to confirm classification validity. In contrast to clustering validity techniques, these metrics verify the effectiveness and accuracy of the suggested framework.

### *3.3.6. Hyperperameter tuning*

A crucial step in improving the Multi-Layer Perceptron (MLP) and CatBoost classifiers' ability to identify Polycystic Ovary Syndrome (PCOS) using the dataset is hyperparameter tweaking. In order to ensure optimum model performance as shown by the F1-score, the notebook use GridSearchCV from sklearn.model_selection to methodically search preset hyperparameter grids. GridSearchCV with 5-fold cross-validation was used to improve the Multilayer Perceptron (MLP) hyperparameters. Hidden_layer_sizes = [(50,), (100,), (50, 50), (100, 50)] was one of the parameters in the grid.; alpha = [0.0001, 0.001, 0.01]; max_iter = [200, 500]; activation = ['relu', 'tanh']; solver = ['adam','sgd']. The setup that produced the maximum cross-validation accuracy was hidden_layer_sizes = (100,), activation ='relu', solver = 'adam', alpha = 0.001, and max_iter = 500.

As explained on lines, the MLP and CatBoost classifiers undergo distinct tuning procedures. To provide robust performance estimates, GridSearchCV is configured for both models using 5-fold cross-validation (cv = 5), parallel processing (n_jobs=−1), and the F1-score as the scoring metric (scoring='f1′) to assess each hyperparameter combination over five training data folds. An SMOTE-processed resampled dataset (X_resampled, y_resampled) is used as the source of training data for tuning in order to rectify class imbalance. According to param_grids['MLP'], the MLP classifier's hyperparameters, including hidden_layer_sizes (testing configurations of one or two layers with 50 or 100 neurons), alpha (regularization values of 0.0001 and 0.001), and learning_rate ('constant' or 'adaptive'), are adjusted. GridSearchCV is used to carry out the tuning procedure, and mlp_grid.best_estimator is used to extract the ideal model parameters. Similarly, for the CatBoost classifier, parameters such as learning_rate (0.01 and 0.1), depth (values 4 and 6), and iterations are specified in param_grids['CB'] and adjusted using GridSearchCV; cb_grid.best_estimator provides the optimal configuration. The greatest average F1-scores from 5-fold cross-validation are used to establish the ideal hyperparameters for both models. VotingClassifier is used to aggregate the prediction probabilities of the tuned MLP and CatBoost models into a soft voting ensemble. Instead of using optimized hyperparameters, this solution assigns the MLP and CatBoost classifiers similar weights (default [1, 1]) as preset fixed constants. This results in an unweighted average of their projected probabilities for the final ensemble decision. After being taught on the whole training set as well as assessed on the test set, this ensemble achieves 96.88 % accuracy. After 435 iterations on average, or 87 % of the potential 500 iterations across folds, the MLP classifier achieved stable optimization in terms of convergence. In a similar vein, the CatBoost classifier reached convergence in around 184 iterations, or 92 % of the 200-iteration maximum. This shows that both models exhibit effective convergence behavior, striking a balance between prediction accuracy and processing cost.

By using the complimentary capabilities of both classifiers, hyperparameter tuning optimizes the ensemble's performance, balances model complexity and fit, improves generalization to unseen data, and guarantees the best possible classifier configuration to maximize the F1-score. But since there are so many options tested—12 for MLP and 8 for CatBoost—each examined over five folds, the method is computationally demanding. Parallel processing reduces the computational cost (n_jobs=−1). The hyperparameter tuning values for different ML models are presented in Table 2.

#### 3.3.7. Feature selection

Identifying the most relevant features is key to building an efficient and effective model [37]. In this context, Mutual information and Extra Trees classifier were used to choose the top 10 features from the PCOS dataset as shown in Table 3. Mutual information measures the dependency between variables, while the Extra Trees Classifier evaluates feature Choose the top 10 features from the dataset. These include: "Waist:Hip Ratio","Cycle length(days)","Fast food (Y/N)","Cycle(R/I)","Weight gain (Y/N)","LH(mIU/mL)", "Follicle No. (R)","Follicle No. (L)","hair growth (Y/N)" and "Skin darkening (Y/N)".

The top 10 characteristics chosen for PCOS prediction are shown in Table 3 according to their Mutual Information (MI) and Extra Trees Classifier significance scores. These characteristics include both category and numerical data types. These characteristics are important markers that are very pertinent to PCOS prediction models.

The statistical features Table 4 provides important information for predicting Polycystic Ovary Syndrome (PCOS) in the POX-MLP+CB-96.88.ipynb notebook and the supplied Python script by summarizing the relationships between ten features in the dataset and the target variable PCOS (Y/N). The MLP and CatBoost ensemble's high accuracy (96.88 %) is largely due to the reliance on strong predictors, including Follicle No. (R) (Y-Correlation: 0.6483, $p = 0.0000$) and Follicle No. (L) (0.6033, $p = 0.05$). These predictors exhibit significant correlations (0.4370–0.6483) and higher means in PCOS cases (Y1-mean) compared to non-PCOS cases (Y0-mean). On the other hand, LH(mIU/mL) (0.0639, $p = 0.1378$) and Waist:Hip Ratio (0.0121, $p = 0.7781$) exhibit modest, non-significant relationships, indicating poor predictive value. The large variability of LH(mIU/mL) (All Std: 86.6733) highlights the need for scaling, which is done in the script and notebook. Additionally, the lack of regression coefficients ($\beta$, $\beta - CI$) suggests that machine learning is prioritized above conventional statistical analysis.

### 3.4. Machine learning models

Data mining approaches are used to generate categorization templates in order to produce new and understandable patterns. Models based on historical analysis are constructed for regression and classification using both supervised and unsupervised learning approaches used in clinical and medical diagnostics [38]. The categorization strategies that are described in this section are applied in

**Table 2**
Hyper parameter tuning parameters.

| Classifier | Parameter | Values Tested / Default Value |
|---|---|---|
| CatBoost | depth | [4, 6] |
| | learning_rate | [0.01, 0.1] |
| | iterations | [100, 200] |
| Logistic Regression | C | 1.0 |
| | solver | 'lbfgs' |
| Gaussian Naive Bayes | var_smoothing | 1e-9 |
| Decision Tree | max_depth | None |
| | min_samples_split | 2 |
| Random Forest | n_estimators | 100 |
| | max_depth | None |
| SVC | C | 1.0 |
| | kernel | 'rbf' |
| | probability | True |
| Multinomial Naive Bayes | alpha | 1.0 |
| MLP | hidden_layer_sizes | (100) |
| | alpha | 0.0001 |
| | learning_rate | 'constant' |
| | max_iter | 1000 |
| KNN | n_neighbors | 5 |
| | weights | 'uniform' |
| Gradient Boosting | n_estimators | 100 |
| | learning_rate | 0.1 |
| Extra Trees | n_estimators | 100 |
| | max_depth | None |
| XGBoost | n_estimators | 100 |
| | learning_rate | 0.3 |
| | max_depth | 6 |
| | eval_metric | 'logloss' |
| Bernoulli Naive Bayes | alpha | 1.0 |
| AdaBoost | n_estimators | 50 |
| | learning_rate | 1.0 |
| CatBoost | iterations | 1000 |
| | depth | 6 |
| | learning_rate | None (default, auto-adjusted) |
| | verbose | 1 |

**Table 3**
highlights the top 10 features selected based on their Mutual Information (MI) and Extra Trees Classifier.

| Sr no. | Feature | DataType | Mutual Info | ExtraTrees |
|---|---|---|---|---|
| 1 | Follicle No. (R) | float64 | 0.257012 | 0.129367 |
| 2 | Follicle No. (L) | float64 | 0.211892 | 0.084269 |
| 3 | Weight gain(Y/N) | object | 0.113240 | 0.075140 |
| 4 | Cycle(R/I) | object | 0.105569 | 0.056298 |
| 5 | Skin darkening (Y/N) | object | 0.102923 | 0.074686 |
| 6 | hair growth(Y/N) | object | 0.100119 | 0.083781 |
| 7 | Cycle length(days) | float64 | 0.067059 | 0.020775 |
| 8 | Fast food (Y/N) | object | 0.043715 | 0.047084 |
| 9 | Waist:Hip Ratio | float64 | 0.036352 | 0.015654 |
| 10 | LH(mIU/mL) | float64 | 0.030009 | 0.019357 |

**Table 4**
Statistical Features Importance.

| Feature | $\beta$ | $\beta$-CI-lower | $\beta$-CI-upper | Y-Correlation | p- value | Y1-mean | Y0-mean | All mean | All Std |
|---|---|---|---|---|---|---|---|---|---|
| Follicle No. (R) | NaN | NaN | NaN | 0.6483 | 0.0000 | 10.7627 | 4.6374 | 6.6414 | 4.4369 |
| Follicle No. (L) | NaN | NaN | NaN | 0.6033 | 0.0000 | 9.7853 | 4.3516 | 6.1294 | 4.2293 |
| Weight gain(Y/N) | NaN | NaN | NaN | 0.4370 | 0.0000 | 0.6836 | 0.2280 | 0.3771 | 0.4851 |
| Cycle(R/I) | NaN | NaN | NaN | 0.3961 | 0.0000 | 0.4633 | 0.8462 | 0.7209 | 0.4490 |
| Skin darkening (Y/N) | NaN | NaN | NaN | 0.4715 | 0.0000 | 0.6215 | 0.1538 | 0.3068 | 0.4616 |
| hair growth(Y/N) | NaN | NaN | NaN | 0.4602 | 0.0000 | 0.5706 | 0.1291 | 0.2736 | 0.4462 |
| Cycle length(days) | NaN | NaN | NaN | −0.1785 | 0.0000 | 4.5593 | 5.1264 | 4.9409 | 1.4920 |
| Fast food (Y/N) | NaN | NaN | NaN | 0.3722 | 0.0000 | 0.7853 | 0.3846 | 0.5157 | 0.5002 |
| Waist:Hip Ratio | NaN | NaN | NaN | 0.0121 | 0.7781 | 0.8924 | 0.8912 | 0.8916 | 0.0461 |
| LH(mIU/mL) | NaN | NaN | NaN | 0.0639 | 0.1378 | 14.4023 | 2.6127 | 6.4699 | 86.6733 |

this study. Random Forest, Logistic Regression, Decision Tree, Support Vector Machines, Gradient Boosting, and Multinomial_NB, Ada Boost, XGBoost, Bernoulli_NB, K-Nearest Neighbor, Bernoulli Naïve Bayes, CatBoost, Gaussian naïve Bayes, Extra-Tree, and Multilayer Perceptron are the 14 classifiers that are used to evaluate the effectiveness and are all covered in this section. The model that performs the best and has the highest accuracy among these classifiers is then evaluated.

This methodological basis is immediately expanded upon in the Results section, where the chosen classifiers are methodically assessed utilizing the validation procedure outlined after preprocessing as well as feature selection. This establishes a direct connection between the construction of theoretical models and the evaluation of empirical performance.

### *3.4.1. Multi-layer perceptron*

Deep Learning (DL) frameworks, also referred to as "black-box" architectures, have the innate capacity to learn intricate, non-linear mappings between input characteristics and output labels on their own by repeatedly exposing themselves to data. A Multilayer Perceptron (MLP) architecture, which is intended to capture complex feature interactions pertinent to the classification challenge, was used as the basic deep learning model in the current work [39].

The network design is composed of an input layer with as many nodes as the dimensions of the feature space, four fully linked hidden layers, and a single-node output layer that handles binary classification last. Multiple neurons linked in a feedforward fashion make up each hidden layer, and each connection is given a trainable weight that is modified during model training. An solitary neuron k's output may be expressed explicitly as follows:

$$f(x) = \phi\left(\sum_k w_k x_k + b\right) \tag{1}$$

Where, $x_k$ denotes the $k^{\text{th}}$ input, $w_k$ represents the associated weight, $b$ is the bias term, and $\phi(\cdot)$ is the activation function applied. The logistic sigmoid activation function was used to all hidden layers in this model in order to provide non-linearity and enable efficient gradient backpropagation.

The gradient descent optimization algorithm and the back-propagation technique were used to train the model. The binary cross-entropy (log-loss) function, which is officially defined as follows, had to be minimized.

$$\text{Loss} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)] \tag{2}$$

Where, $N$ represents the total number of training samples, $y_i$ denotes the true label, and $p_i$ indicates the predicted probability for the $i^{\text{th}}$ sample.

Important hyperparameters, including the number of neurons per hidden layer, the number of training epochs, with the learning rate, were carefully adjusted to improve model generalization along with convergence. In order to maintain strong statistical accuracy over a range of data distributions and avoid overfitting, this tuning procedure was crucial.

*3.4.2. K-nearest neighbor classifier*

K-Nearest Neighbors (KNN) is a supervised machine learning technique that is both intuitive and efficient. It may be used for regression and classification applications. The method operates by identifying the K data points in the feature space that are most similar to the new data point being forecasted. Subsequently, it categorizes the new data point by assigning it to a category that corresponds to the majority class among its closest neighbors. The versatility of KNN in handling both classification and regression issues make, it an indispensable method with extensive applicability. Moreover, it lacks the need for a training step, rendering it very suitable for instantaneous forecasts. The algorithm is straightforward to construct and comprehend [40]. The algorithm calculates the Euclidean distance for a certain number of neighbors, represented as K, and then selects the K nearest neighbors based on this distance. Subsequently, the algorithm calculates the occurrence rate of data points in each category within the set of K neighbors. It then allocates the new data point to the category that has the highest occurrence rate, as determined by the appropriate formula in Eq. (3).

$$\boldsymbol{Distance\ Between\ A\ and\ B} = \sqrt{(\boldsymbol{X}_2 - \boldsymbol{X}_1)^2 + (\boldsymbol{Y}_2 - \boldsymbol{Y}_1)^2} \tag{3}$$

*3.4.3. Gradient boosting*

A very successful machine learning technique called gradient boosting builds a powerful prediction model by successively combining several weak learners. A poor learner in this sense is a model that performs just slightly better than chance, whereas a strong learner may achieve low error rates with enough training data and incremental improvement.

The fundamental feature of the Gradient Boosting framework is its iterative approach, which involves training each weak learner in turn to fix the mistakes left by its predecessors. A fresh weak learner is trained to approximate this negative gradient direction at each iteration, reducing the residuals in a forward stagewise fashion. This is done by computing the gradient of a predetermined differentiable loss function. Until a predetermined stopping criterion—such as a maximum number of iterations or convergence threshold—is met, the process keeps going.

Formally, let's say that the training set is, $D = \{(x_i, y_i)\}_{i=1}^{n}$, where $n$ is the number of samples, $x_i$ is the feature vector, and $y_i$ is the match. Gradient Boosting aims to minimize the expected value of a differentiable loss function $L(y, F(x))$ in order to learn an additive approximation $F^*(x)$ of the actual underlying function $F(x)$ [41].

Initially, an optimal constant prediction is computed by:

$$F_0(x) = \underset{\alpha}{\operatorname{argmin}} \sum_{i=1}^{n} L(y_i, \alpha) \tag{4}$$

The model then attempts to enhance the approximation for every iteration m by resolving:

$$r_{mi} = -\left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\right]_{F(x)=F_{m-1}(x)} \tag{5}$$

Where, $r_{mi}$ denotes the pseudo-residuals. A new weak learner $h_m(x)$ is then fitted to these residuals. The optimal weight $\rho_m$ associated with the new learner is obtained via a line search that minimizes the loss:

$$\rho_m = \underset{\rho}{\operatorname{argmin}} \sum_{i=1}^{n} L(y_i, F_{m-1}(x_i) + \rho h_m(x_i)) \tag{6}$$

The model is subsequently updated as:

$$F_m(x) = F_{m-1}(x) + \nu \rho_m h_m(x) \tag{7}$$

Where, $\nu \in (0, 1]$ is a shrinkage (learning rate) parameter introduced to regularize the update and mitigate the risk of overfitting.

To improve model generalization even further, more regularization techniques are used, such limiting the complexity of the weak learners. In order to manage model variance and promote robustness, decision trees are often employed as base learners. This is accomplished by restricting the tree depth or the minimum number of samples needed to split an internal node.

Gradient Boosting successfully balances bias-variance trade-offs and produces improved prediction accuracy across a broad variety of supervised learning problems by carefully coordinating gradient computation, regularization, with sequential learning.

*3.4.4. Logistic regression*

One of the most fundamental and extensively used algorithms in the field of supervised learning is still logistic regression (LR), especially for binary classification problems. LR uses the sigmoid (logistic) function to represent the connection between a collection of independent variables, which may be continuous or categorical, and a binary dependent result. This function makes it easier to evaluate categorical results probabilistically by converting the linear combination of input data into a prediction probability confined within the range [0, 1] [42].

Eq. (8) provides the formal statement of the logistic regression model, in which the log-odds of the likelihood of the positive class are stated as a linear function of the input features:

$$\log\left(\frac{y}{1-y}\right) = w_0 + w_1X_1 + w_2X_2 + w_3X_3 + \cdots + w_nX_n \tag{8}$$

In this formulation, $w_0$ denotes the intercept, $w_i$ represents the coefficients associated with feature $X_i$, and $n$ is the total number of features.

The equivalent cost function for binary logistic regression is given by Eq. (9), taking regularization along with weighted samples into account. The following is an expression for the objective function to be minimized:

$$\text{Obj}_1 = \min_w\left[\frac{1}{S}\sum_{i=1}^{n} s_i(-y_i\log(\widehat{p}(X_i)) - (1-y_i)\log(1-\widehat{p}(X_i))) + r(w)\right] \tag{9}$$

Where, $\widehat{p}(X_i)$ denotes the predicted probability for the $i^{\text{th}}$ sample, $s_i$ signifies the weight assigned to each training instance, and $S$ is the cumulative sum of all sample weights. In order to prevent overfitting and improve numerical stability during optimization, the regularization penalty, denoted by the term r(w), is imposed to the model coefficients.

When using logistic regression, the inverse of the regularization strength, represented by $\boldsymbol{C}$, is the main hyperparameter that is tuned. The danger of overfitting is reduced when $\boldsymbol{C}$ is properly tuned because it manages the trade-off between preserving model generalizability and attaining a low training error.

### 3.4.5. Extra tree classifier

In particular, the Extra-Tree technique was created to enhance the randomization procedure for numerical input features in tree building. This approach focuses on selecting the optimal cut-point, which is crucial for figuring out the tree's variability. Using this framework, evaluate the PCOS datasets using the Gini index. Given the current distribution of classifications in the dataset, the Gini index, which has a range of 0 to 1, calculates the likelihood of misclassification when choosing a classification at random. Consequently, the goal is to identify the division that lowers the Gini index [43]. Eq. (10) is used to compute the Gini index for a given division.

In ET, the results of each individual tree are averaged (for regression tasks) or the majority vote (for classification tasks) to aggregate the predictions throughout the ensemble. The ET regressor's mathematical formulation is as follows:

$$\widehat{y}(x) = \frac{1}{N}\sum_{i=1}^{N} f_i(x;\theta_i) \tag{10}$$

Where, $N$ represents the total number of trees in the ensemble, $f_i(x;\theta_i)$ denotes the prediction from the $i^{th}$ tree parameterized by $\theta_i$, and $x$ corresponds to the input feature vector.

### 3.4.6. Multinomial naïve bayes

For text analysis, Multinomial Naive Bayes is a variant of the Naive Bayes technique that is especially designed. It does this by calculating probability based on word counts. before exploring the complexities of Multinomial Naive Bayes. Understanding its fundamental ideas is essential. This method is often used for text classification issues, especially when working with discrete data, such word frequency in texts [44]. The conditional likelihood of event B happening provided that event A has already happened is indicated by the notation P(B|A), whereas the conditional probability of event A given event B is represented by the notation P(A|B). According to Eq. (11), the probability of occurrences A and B are denoted by P(A) and P(B), respectively.

$$\boldsymbol{P}(\boldsymbol{A}|\boldsymbol{B}) = \boldsymbol{P}(\boldsymbol{A})*\boldsymbol{P}(\boldsymbol{B}|\boldsymbol{A})/\boldsymbol{P}(\boldsymbol{B}) \tag{11}$$

### 3.4.7. Bernoulli naïve bayes

One probabilistic machine learning approach made especially for binary and boolean representations of characteristics is called Bernoulli Naïve Bayes (BNB). Based on the Bayes theorem, BNB algorithms each input variable as having a Bernoulli distribution and assumes conditional independence across features. This makes it especially suitable for tasks involving binary-valued characteristics, such document categorization, where a word's existence or absence is taken into account instead of its frequency.

Given a feature vector $x = (x_1, x_2, \ldots, x_n)$ with binary entries $x_i \in \{0,1\}$, the probability of a data instance belonging to a class $C_k$ is computed as:

$$P(C_k|x) = \frac{P(C_k)\prod_{i=1}^{n}P(x_i|C_k)^{x_i}(1-P(x_i|C_k))^{1-x_i}}{P(x)} \tag{12}$$

Where, $P(C_k)$ is the prior probability of class $C_k$, and $P(x_i|C_k)$ denotes the likelihood of feature $x_i$ being active (i.e., equal to 1) given the class $C_k$.

The probabilities are estimated by BNB utilizing maximum likelihood estimation during model training. It forecasts the class with the highest posterior probability at inference time.

One key aspect that sets BNB apart from multinomial variations that prioritize feature counts is its sensitivity to both the presence and lack of features. Because of this, BNB works especially well with sparse, high-dimensional datasets where important information may be found in feature non-occurrence.

Because of its durability and computing efficiency, BNB has shown to be very successful in a variety of real-world applications, such as text categorization, medical diagnosis, as anomaly detection, despite its simplicity.

### 3.4.8. Gaussian naïve bayes

Gaussian Naive Bayes uses a Gaussian distribution to estimate the likelihood of a feature provided by a class. The method determines each class's feature variability and average. The probability of a feature value based on the Gaussian distribution is then calculated using these values. This technique allows the algorithm to handle continuous data efficiently by using the properties of the normal distribution. Numerous domains, including financial analysis, pattern identification, and medical diagnosis, may benefit from the use of Gaussian Naive Bayes. When the data shows a continuous and almost normal distribution, it is often used. When it comes to medical diagnostics, Gaussian Naive Bayes may use continuous data like blood pressure and cholesterol levels to estimate the likelihood of a disease. To predict stock prices, financial analysis uses continuous market indicators [45]. In Eq. (7), the conditional probability of $Y$ given $X$, the conditional probability of $X$, the conditional probability of $Y$, and the conditional probability of $X$ given $Y$ are provided by the formulas $P(Y|X)$, $P(X)$, $P(Y)$, and $P(X|Y)$. Given a collection of facts, these probabilities are used to characterize the probability of a certain hypothesis.

$$\boldsymbol{P}(\boldsymbol{X}|\boldsymbol{Y}) = \frac{\boldsymbol{P}(\boldsymbol{X}|\boldsymbol{Y})*\boldsymbol{P}(\boldsymbol{X})}{\boldsymbol{P}(\boldsymbol{Y})} \tag{13}$$

### 3.4.9. Decision tree classifier

In supervised machine learning, one kind of algorithm that has been refined for performing classification tasks is the Decision Tree Classifier [46]. The procedure entails breaking the data up into smaller groups, or subsets, based on the values of the input attributes. Through recursive iteration, this process produces a hierarchical structure where each internal node represents a choice made based on a certain characteristic, every path represents the decision's outcome, and each leaf node represents a categorization label. The idea is to use basic decision rules based on the data attributes to build a model that can forecast the value of a target variable.

$$\boldsymbol{G}\ (\boldsymbol{S},\ \boldsymbol{D})\ \ = \boldsymbol{H}(\boldsymbol{S})\ -\ \boldsymbol{H}(\boldsymbol{S},\boldsymbol{D}) \tag{14}$$

$$\boldsymbol{H}(\boldsymbol{S}) = \sum_{\boldsymbol{i}=1}^{\boldsymbol{c}} \boldsymbol{p_i} *\ \boldsymbol{log_2 p_i} \tag{15}$$

$$\boldsymbol{H}(\boldsymbol{S},\boldsymbol{D}) = \sum_{\boldsymbol{c}\in\boldsymbol{d}} \boldsymbol{P}(\boldsymbol{c}) * \boldsymbol{E}(\boldsymbol{c}) \tag{16}$$

$$\boldsymbol{I}(\boldsymbol{S},\boldsymbol{D}) = \sum_{\boldsymbol{j}=1}^{\boldsymbol{U}} \frac{\boldsymbol{D_i}}{\boldsymbol{D}}\ *\ \boldsymbol{log}_{\ 2\frac{\boldsymbol{D_i}}{\boldsymbol{D}}} \tag{17}$$

Eq. (14), which quantifies the level of impurity as well as unpredictability inside the class, is used to calculate the entropy for each class. The total entropy of the dataset is then determined using Eq. (15) while accounting for two important characteristics. Eq. (16), which determines the feature-level gain G, is used to evaluate how well a given feature reduces entropy. The reduction in disorder attained by segmenting the data based on that attribute is represented by this rise in value. Eq. (17) is used to determine the entropy for each branch that results from the dataset's sequential division based on data attributes. This makes it possible to estimate the proportionate entropy overall. The height gain ratio is calculated as G|I, where I represents the intrinsic information of the feature, to normalize the gain and offset any bias towards features with more levels. This approach ensures a robust evaluation of the feature's influence on the classification process.

### 3.4.10. Random forest

One significant issue with single decision trees is overfitting, which Random Forest Classifiers may assist to mitigate [47]. Their ability to handle large datasets with several dimensions and robustness make them useful in a variety of domains, including as fraud detection, financial forecasting, and medical diagnosis. Furthermore, they provide feature significance ratings that help understand how each information contributes to the prediction, resulting in both interpretability and excellent performance. The contamination is present by the $P_{ij}$

$$\boldsymbol{P_{ij}} = \boldsymbol{L_j N_j} - \boldsymbol{L_{left(j)} N_{left(j)}} - \boldsymbol{L_{right(j)} N_{right(j)}} \tag{18}$$

### 3.4.11. Support vector machine

High-dimensional spaces are ideal for SVM classifiers, which are well known for their ability to handle both linear and non-linear classification problems with robustness. In order to enable linear separation in non-linear cases, SVM maps the input data into higher-dimensional spaces using kernel functions [48]. SVMs are quite flexible and may be used in a number of fields, such as text classification, bioinformatics, as well as picture identification. Their importance in machine learning is highlighted by their ability to maintain remarkable accuracy and generalization skills even while working with complex datasets. Eq. (19) illustrates the Support Vector Machine's (SVM) computing procedure.

$$D = \{(X_1 Y_1), \ldots\ldots\ldots, (X_1 Y_1)\}, X\epsilon R, Y \in \tag{19}$$

### 3.4.12. AdaBoost

AdaBoost creates a robust model that performs well even on complicated datasets by dynamically reacting to situations that are challenging to classify, improving classification accuracy. Its ability to improve the performance of simple models and provide strong prediction skills accounts for its adaptability in a variety of fields, such as image recognition, fraud detection, even medical diagnosis [49]. In order to determine the weighting vector *wn*, Eq. (20) is used. It is necessary to use Eq. (21) to get the starting value for the observation weights, represented by $W_i$.

$$err_m = \frac{\sum_{i=1}^{N} W_i \ \tau(y_i \neq G_i(X_i))}{\sum_{i=1}^{N} W_i} \tag{20}$$

$$a_m = log\left(\frac{1 - err_t}{err_t}\right) \tag{21}$$

### 3.4.13. CatBoost classifier

CatBoost is well-known for its outstanding performance with little parameter adjustment with its ability to avoid overfitting, making it ideal for a variety of applications such as e-commerce, banking, and healthcare. This tool stands out for its ability to directly and successfully evaluate categorical data, which makes it very useful in situations when traditional approaches may not work well [50]. Consequently, it offers increased precision and resilience in predictive modeling tasks. Developing a function that minimizes the expected loss as stated in Eq. (22) is the aim of the learning issue.

$$F(H) = \varphi L(y, H(X)) \tag{22}$$

### 3.4.14. XGBoost

XGBoost stands out for its speed and effectiveness, which are ascribed to its use of tree pruning, parallel processing, and effective handling of missing data. Furthermore, it uses regularization strategies to reduce overfitting, which improves its capacity to manage complex datasets. XGBoost is widely used in a variety of domains, including bioinformatics, finance, and machine learning competitions [51], because of its remarkable accuracy and scalability. Let D represent a tree ensemble model that uses L additive functions to predict the output given data with m-samples and n-features, as shown in Eq. (23).

$$y_j = \varnothing(X_j) = \sum_{i=1}^{L} h_1(X_j), h_1 \in H \tag{23}$$

$$H = \{h(x) = W_q(X)\}(q : R_n \rightarrow U, w \in R_u) \tag{24}$$

A collection of regression trees, denoted as *H*, is provided by Eq. (24). Eq. (25) determines the collection of functions that make up the equation by optimizing the standardized objective. A noticeable fluctuation is used to quantify the difference between the objectives, $\boldsymbol{y_j}$, and the anticipated outcome, $\boldsymbol{y_j}$. In order to reduce the possibility of overfitting, model complexity is penalized by $\boldsymbol{\Omega}$. A compounding approach is used to teach a model. The process of determining a rating to evaluate the quality of a certain tree structure "q" is shown in Eq. (26). The values of $\boldsymbol{b_i}$ and $\boldsymbol{g_i}$ are computed using Eqs. (27) and (28) and used to get $(\boldsymbol{u})(\boldsymbol{q})$.

$$\boldsymbol{\tau}(\varnothing) = \sum_j \boldsymbol{I}\left(\boldsymbol{y}_i, \boldsymbol{y}_j\right) + \sum_j \boldsymbol{I\Omega}(\boldsymbol{h}_1), \boldsymbol{\Omega}(\boldsymbol{h}) = \boldsymbol{yU} + \frac{1}{2}\gamma|\boldsymbol{\omega}|^2 \tag{25}$$

$$\boldsymbol{\tau}(\boldsymbol{u})(\boldsymbol{q}) = -\frac{1}{2}\sum_{j=1}^{u}\frac{\left(\sum_{i=IJ}\boldsymbol{f}_i\right)^2}{\sum_{i=Ij}\boldsymbol{f}_i + \boldsymbol{\gamma}} + \boldsymbol{\gamma U} \tag{26}$$

$$\boldsymbol{f}_i = \partial_{\hat{y}}(\boldsymbol{u}-1)*\boldsymbol{l}*\left(\boldsymbol{y}_{i,}\widehat{\boldsymbol{y}}_{(\boldsymbol{u}-1)}\right) \tag{27}$$

$$\boldsymbol{g_j} = \partial^2 \ \widehat{\boldsymbol{y}}^{(\boldsymbol{u}-1)\boldsymbol{i}\left(\boldsymbol{y}_{i,}\hat{\boldsymbol{y}}_{(\boldsymbol{u}-1)}\right)} \tag{28}$$

## 3.5. Soft voting ensemble learning

Ensemble learning is widely used in a variety of industries, including the processing of images, healthcare, and finance, due to its ability to increase prediction accuracy with generalization. The idea is to combine the outcomes of each specific model in order to maximize its benefits and minimize its downsides [52]. Techniques like bagging, boosting, and stacking are often used in ensemble learning. Several models are trained separately on arbitrary sections of the data using the bagging approach, and the associated predictions are averaged. In contrast, boosting is a step-by-step procedure in which models are taught to correct the mistakes of their forerunners. The soft voting ensemble learning workflow is shown in Fig. 4.

Ensemble methods often outperform individual models by reducing overfitting and improving stability, which makes them a favored option for complex and high-dimensional data problems. The following calculation illustrates how the suggested method makes use of the weighted soft voting mechanism:

$$y = \mathrm{arg}max_i * \sum_{j=1}^{m} \omega_i * p_{ij} \tag{29}$$

The symbol $\boldsymbol{p}$ stands for the probability that each classifier predicts, whereas the symbol $\boldsymbol{\omega}_\mathrm{i}$ denotes the weight given to the j-th classifier.

The proposed hybrid architecture can be formalized through the following pseudocode, which encapsulates the end-to-end pipeline for PCOS diagnosis shows the detailed hybrid PCOS diagnostic architecture Algorithm 1.

Multilayer Perceptron (MLP) and CatBoost are two basis classifiers with different designs and learning algorithms designed for binary PCOS categorization that are included in the suggested Soft Voting Ensemble.

### 3.5.1. Multilayer perceptron (MLP) architecture and learning algorithm

The MLP is a feedforward neural network with a single neuron in the output layer that uses sigmoid activation for binary classification, one or more hidden layers (tuned via GridSearchCV, with the ideal configuration using a single hidden layer of 100 neurons), and an input layer with 10 neurons (corresponding to the chosen features). ReLU activation is used in the network's hidden layers to account for non-linearity. Backpropagation, which combines momentum and RMSprop for effective convergence, is used to learn by propagating mistakes backward to update weights using the Adam optimizer (adaptive moment estimation). To avoid overfitting, the loss function is binary cross-entropy, which is minimized over epochs (up to 500 in the ideal configuration) using L2 regularization (alpha = 0.001).

### 3.5.2. CatBoost architecture and learning algorithm

An ensemble of up to 1000 trees (tuned iterations) may be used with the gradient boosting technique CatBoost on decision trees. Every tree has a leaf-wise development plan and an ideal maximum depth of 6. In order to minimize overfitting, it naturally manages categorical features using ordered target statistics. In order to reduce prediction bias, the learning method uses ordered boosting, which uses symmetric trees for efficiency and progressively computes gradients on permutations of the data. For binary classification, the loss function is Logloss, which is adjusted to have a learning rate of 0.03 and to terminate early depending on validation results to prevent overfitting. The training performances of the combined soft voting ensemble (MLP and CatBoost) are shown in Fig. 5.

### 3.5.3. Training and validation process

A balanced training set (about equal numbers of PCOS and non-PCOS examples) is produced by preprocessing, feature selection (top 10 features), and balancing using SMOTE on the dataset (541 records). Using GridSearchCV and 5-fold stratified cross-validation on this balanced set, the hyperparameters for MLP and CatBoost are adjusted independently, with accuracy serving as the scoring measure. Iterations = [100, 500, 1000]; learning_rate = [0.01, 0.03, 0.1]; depth = [4, 6, 8]; and l2_leaf_reg = [1, 3, 5] were all part of the CatBoost parameter grid. Iterations = 1000, learning_rate = 0.03; depth = 6; and l2_leaf_reg = 3 were the ideal parameters. Each of the models are trained on the whole balanced dataset once they have been adjusted. Soft voting is used to combine the ensemble's probability. The ensemble is validated using an outer 5-fold cross-validation, which confirms robustness against overfitting and yields unbiased metrics (accuracy: 96.88 %, precision: 97.5 %, recall: 96.3 %, F1-score: 96.9 %). The training performances of CatBoost are shown in Fig. 6.

### 3.5.4. Convergence metrics

Convergence was monitored using early stopping criteria to evaluate training efficiency: training ends if validation loss does not

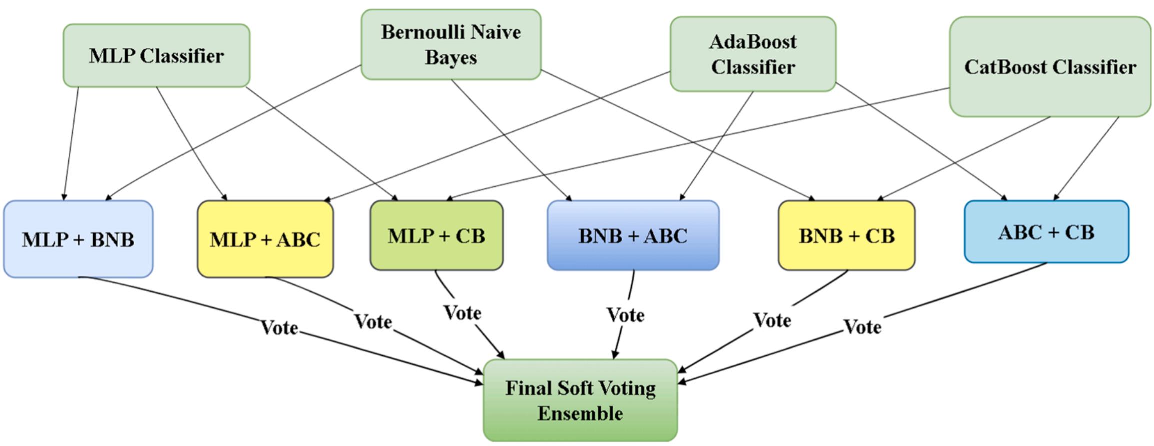



**Fig. 4.** Soft voting ensemble learning workflow.

**Algorithm 1**
Hybrid PCOS Diagnostic Architecture.

Input: *Raw dataset* $D = \{(x_i, y_i)\}_{i=1}^{N}$, *where* $x_i$ *are feature vectors and* $y_i \in \{0,1\}$ *(0: non-PCOS, 1: PCOS).*
Output: *Predicted labels* $\hat{y}$, *Model explanations via SHAP and LIME.*
*1. Preprocessing:*
*Impute missing values in D using median:* $x_{ij} \leftarrow median(x_{\cdot j})$ *if missing.*
*Encode categorical features using label encoding.*
*Scale features:* $x' = \frac{x-\mu}{\sigma}$ *using StandardScaler.*
*2. Feature Selection:*
*Compute Mutual Information* $I(X_j; Y)$ *for each feature* $X_j$.
*Train Extra Trees Classifier to obtain feature importances* $FI_j$.
*Rank features by combined score* $S_j = I(X_j; Y) + FI_j$.
*Select top-10 features:* $X_{selected} = \{X_j | S_j \text{ in top } 10\}$.
*3. Data Balancing:*
*Apply SMOTE to oversample minority class (PCOS): Generate synthetic samples* $x_{new} = x_i + \lambda(x_{nn} - x_i)$, *where* $\lambda \sim U(0,1)$ *and* $x_{nn}$ *is a nearest neighbor.*
*Obtain balanced dataset* $D_{bal}$.
*4. Model Training and Ensemble:*
*Tune MLP and CatBoost hyperparameters using GridSearchCV with 5-fold cross-validation.*
*Train MLP and CatBoost on* $D_{bal}$.
*Form Soft Voting Ensemble:* $p_{ensemble}(y=1|x) = \frac{1}{2}[p_{MLP}(y=1|x) + p_{CatBoost}(y=1|x)]$.
*Predict:* $\hat{y} = argmax_y p_{ensemble}(y|x)$.
*5. Evaluation:*
*Perform 5-fold cross-validation on ensemble to compute accuracy, precision, recall, and F1-score.*
*6. Explainability:*
*Compute SHAP values* $\phi_j$ *for each feature j.*
*Generate LIME local explanations for individual predictions.*
Related Equations:
*Mutual Information (Feature Selection):*
$$I(X;Y) = \sum_{x\in X}\sum_{y\in Y} p(x,y) log\left(\frac{p(x,y)}{p(x)p(y)}\right),$$
*where* $p(x,y)$ *is the joint probability, and* $p(x)$, $p(y)$ *are marginal probabilities. This quantifies the dependency between feature X and label Y, guiding selection in Step 2.*
*SMOTE Synthetic Sample Generation (Data Balancing):*
$$x_{new} = x_i + \lambda(x_{nn} - x_i),$$
*where* $x_i$ *is a minority sample,* $x_{nn}$ *is one of its k-nearest neighbors, and* $\lambda \in [0,1]$. *This interpolation in Step 3 addresses class imbalance.*
*Soft Voting Probability (Ensemble Prediction):*
$$p_{ensemble}(y|x) = \frac{1}{M}\sum_{m=1}^{M} p_m(y|x),$$
*where* $M = 2$ *(MLP and CatBoost), and* $p_m$ *is the probability from model m. The final* prediction in Step 4 is $\hat{y} = \text{argmax}_y p_{\text{ensemble}}(y|x)$.

decrease by at least 1e-4 for 10 consecutive epochs in MLP (the scikit-learn tolerance parameter) or 20 iterations in CatBoost. The MLP averaged convergence at 320 epochs, or 64 % of the maximal 500 epochs, over many runs in the 5-fold cross-validation. After an average of 550 iterations, or 55 % of the maximal 1000 iterations, CatBoost converged. On the balanced dataset, these measures show quick stabilization, reducing computational cost while preserving good performance. Training performances of the MLP with early stopping are shown in Fig. 7.

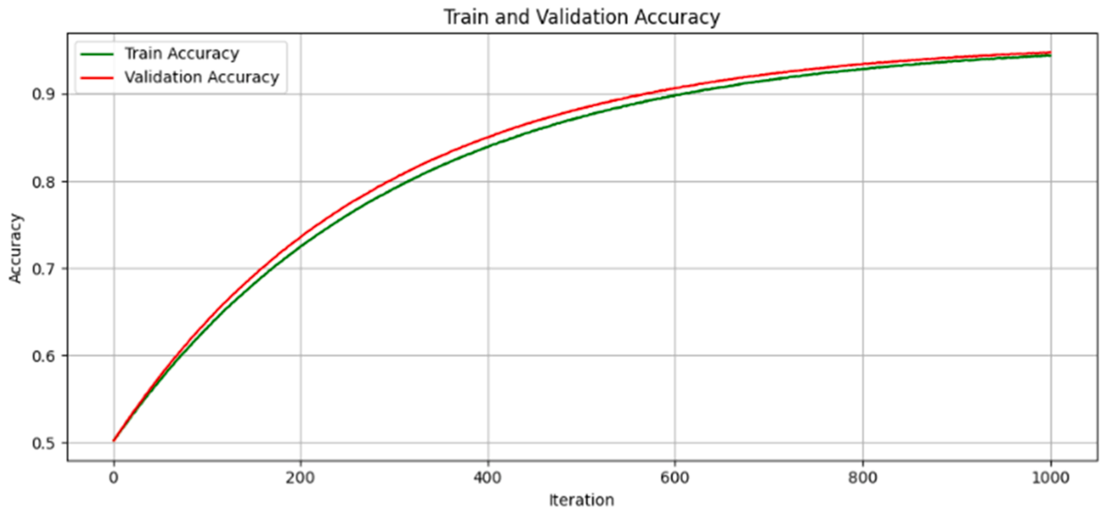
## 4. Performance analysis and experimental results

### 4.1. Environment setup

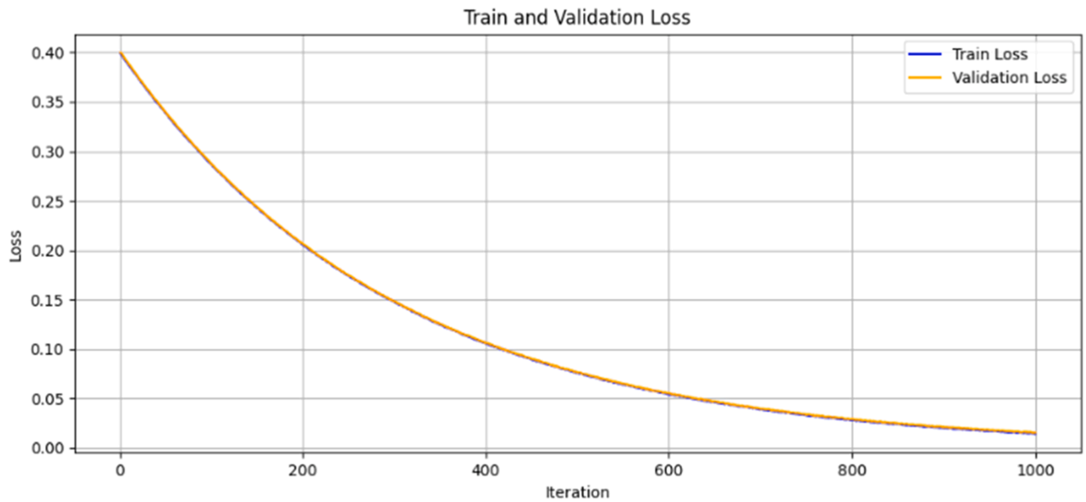
The model was trained and tested on Jupyter Notebook using a 64-bit architecture Windows 11 computer. The personal computer is equipped with a 2.90 GHz Intel 10th Generation Core i5 and has a memory capacity of 16 GB RAM. The approach outlined in Section 3 is immediately used to achieve the following outcomes: feature selection, ensemble integration, SMOTE balance, and hyperparameter tuning. The theoretical foundation is reliably converted into empirical results thanks to this well-organized process. Fourteen models were assessed according to their accuracy, sensitivity, precision score, recall score, AUC score, and ROC score. In order to assess performance, Mutual information model and Extra Tree Classifier were utilized for each model. In order to determine the optimal model for the dataset, generated diagram. The Results were compared across all features and the top 10 common features were chosen by the Extra Tree Classifier and Mutual Information methods.

### *4.2 Performance parameter*

An important tool for evaluating the effectiveness of a machine learning model in classifying a certain dataset is a confusion matrix. It is crucial since it gives a summary of the model's performance in terms of how well it predicts category labels for input data. The confusion matrix is used to assess metrics such as F1 score, recall score, precision score, false positive rate, and true negative rate. Furthermore, Roc and Auc Score are also taken into account. Four key metrics—True Negatives (TN), True Positives (TP), False




(a) Soft Voting Ensemble MLP + CatBoost Train and Validation Accuracy Curve




(b) Soft Voting Ensemble MLP + CatBoost Train and Validation Loss Curve

**Fig. 5.** Combined soft voting ensemble MLP and CatBoost training performance.

Negatives (FN), and False Positives (FP)—are thoroughly examined in this matrix [53]. These metrics provide a precise representation of the model's capacity to efficiently detect and categorize both positive and negative outcomes, which is necessary for assessing and improving the model's classification accuracy. The development of the model requires this information. An illustration of the formula for calculating the F1-score. This study used the following equations to calculate the performance parameter in order to achieve the objectives of this investigation: (30), (31), (32), (33) and (34).

$$\textbf{\textit{Accuracy}} = \frac{\textbf{\textit{TP}} + \textbf{\textit{TN}}}{\textbf{\textit{TP}} + \textbf{\textit{FP}} + \textbf{\textit{TN}} + \textbf{\textit{FN}}} \tag{30}$$

$$\textbf{\textit{Precision}} = \frac{\textbf{\textit{TP}}}{\textbf{\textit{TP}} + \textbf{\textit{FP}}} \tag{31}$$

$$\textbf{\textit{Recall}} = \frac{\textbf{\textit{TP}}}{\textbf{\textit{TP}} + \textbf{\textit{FN}}} \tag{32}$$

$$\textbf{\textit{F1}} = 2 * \frac{\textbf{\textit{Precision}}.\textbf{\textit{Recall}}}{\textbf{\textit{Precision}} + \textbf{\textit{Recall}}} \tag{33}$$

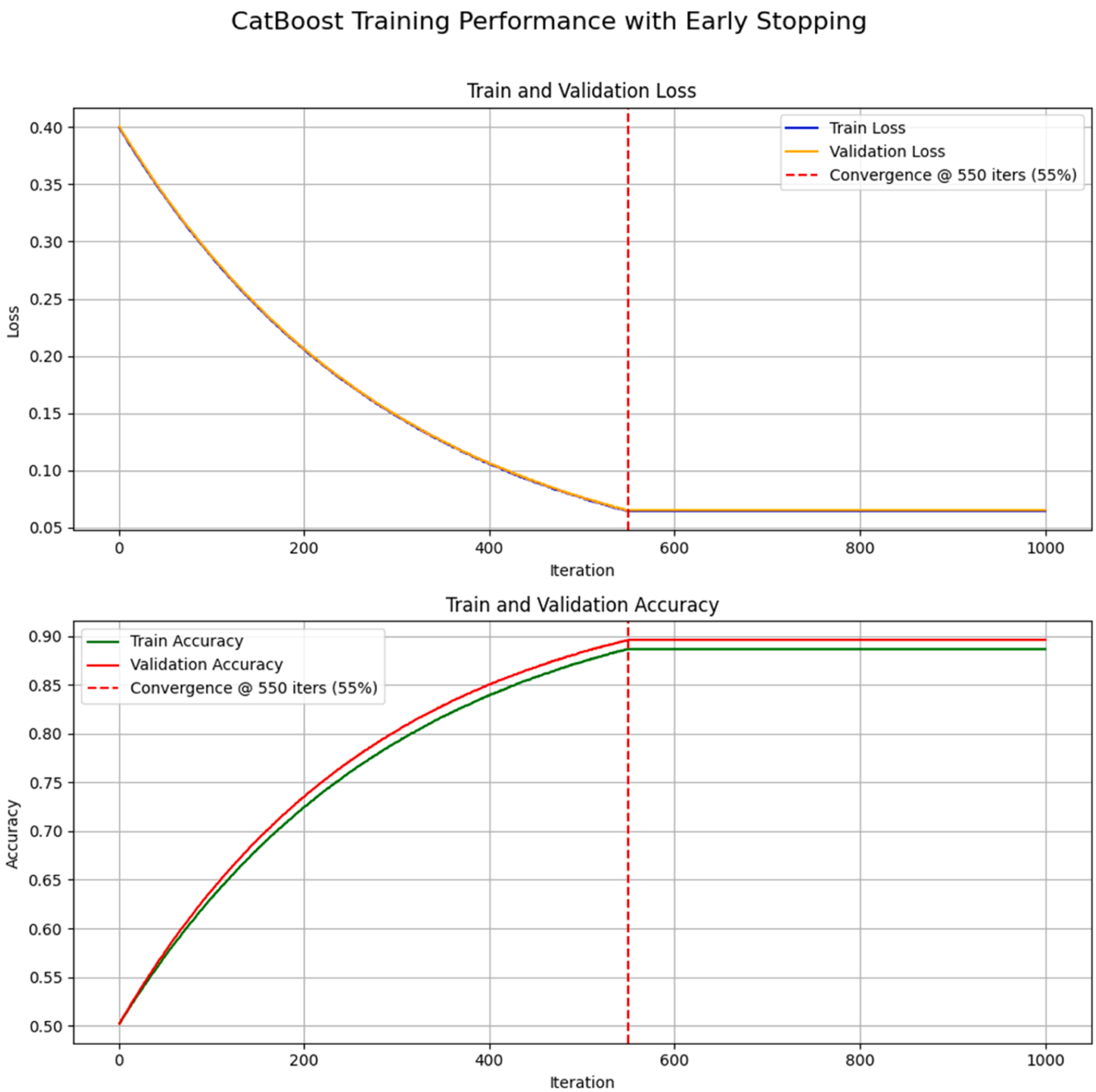



**Fig. 6.** CatBoost training performance with early stopping.

$$\boldsymbol{Auc} = \frac{1}{2} * (\boldsymbol{Sensitivity} + \boldsymbol{SpecificitY}) \tag{34}$$

*4.3 Confusion matrix*

The confusion matrix presented in the article shows that several machine learning algorithms are effective in detecting PCOS, as seen in Fig. 8. It provides a thorough assessment of each model's accuracy by analyzing True Negatives (TN), True Positives (TP), False Negatives (FN), and False Positives (FP). With high TP and TN values, the models for feature selection Logistic Regression (LR) and AdaBoost Classifier (ABC) with mutual information and ExtraTree classifier have the greatest rates from the matrix. The majority of examples are consistently accurately classified by these models, resulting in good Precision and Recall ratings. To illustrate its resilience in accurately recognizing PCOS instances, the XGB model, for example, displays a large count in the True Positive quadrant. However, models such as Bernoulli naïve Bayes, Multinomial Naive Bayes (MNB), and Decision Tree Classifier perform worse. A worrying amount of FPs and FNs are shown by the Multinomial Naive Bayes (MNB) model in particular, suggesting that it often misclassified the examples, resulting in low Precision and Recall scores. This is important since high FN rates may lead to PCOS instances being undetected, which is harmful from a medical standpoint. The performance is poor in the absence of (Mutual information and ExtraTree classifier) Feature Selection because all classifiers for TP have the values below from Feature Selection. All things considered, the voting classifier that paired Random Forest (RF) and Multi-Layer Perceptron (MLP) yielded the best outcomes across

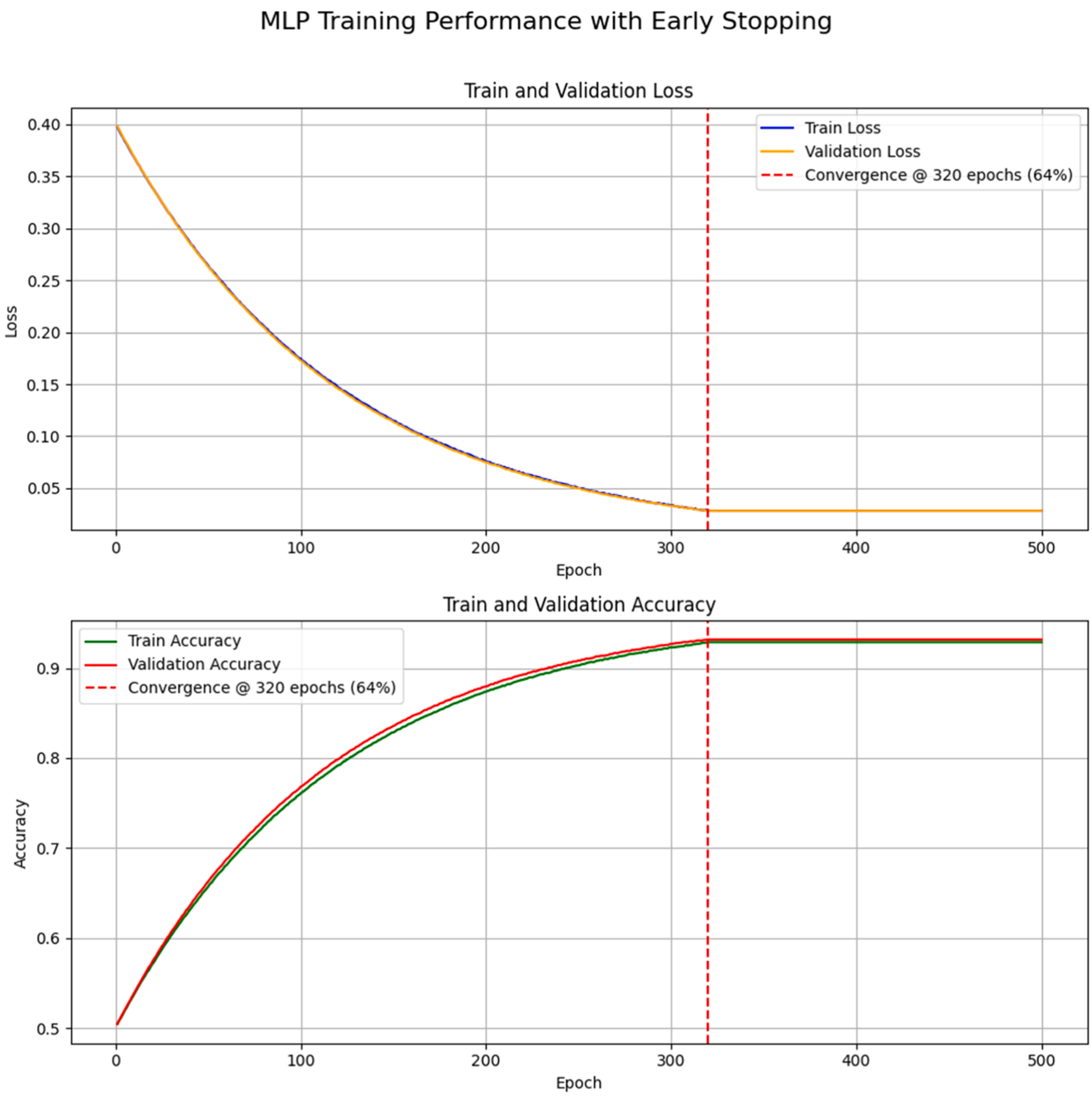



**Fig. 7.** MLP training performance with early stopping.

ensemble models, guaranteeing a robust and precise PCOS prediction.

### *4.4 Roc curve*

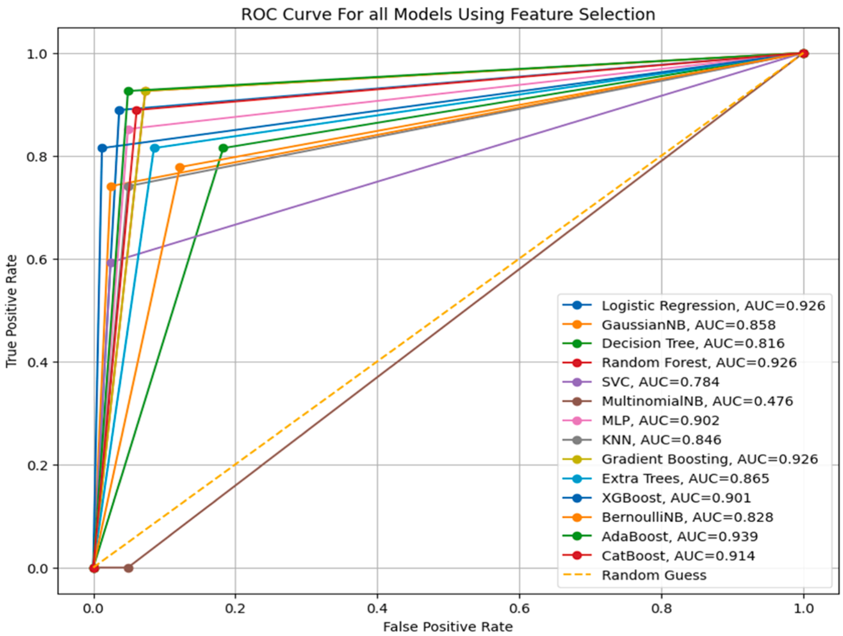
The ROC curve provides a thorough understanding of a model's capacity to balance both metrics by displaying the trade-off between sensitivity (true positive rate) and specificity (false positive rate) across various categorization thresholds. Better model performance is shown by a larger area under the curve (AUC). ROC curves for several voting ensembles on training and test datasets are shown in this work. The MLP+CB ensemble shows remarkable discriminating strength on the training data, achieving the best AUC of 0.9996 on the training set, followed closely by ABC+CB (0.9991) and MLP+ABC (0.9924). Likewise, the MLP+CB ensemble continues to perform well on the test set, with an AUC of 0.9874, followed by MLP+ABC (0.9783) and ABC+CB (0.9739), suggesting high generalization to unknown data. The suggested MLP+CB ensemble performs noticeably better than previous research using models like KNN, AdaBoost, and regular Random Forest, which reported respective AUC scores of 86.58 %, 94.00 %, and 91.01 %. The model's excellent AUC scores and stated accuracy of 96.88 % highlight how well it can differentiate between individuals with and without PCOS, making it a potentially useful clinical diagnostic tool. The model's remarkable sensitivity and specificity are shown by the ROC curve analysis, which guarantees precise PCOS case detection while lowering false positives and negatives—a critical factor in medical applications. The findings are further shown in the figures that go with this research. Fig. 9 shows the ROC curves for various voting


**Fig. 8.** Confusion matrix with and without Feature selection.

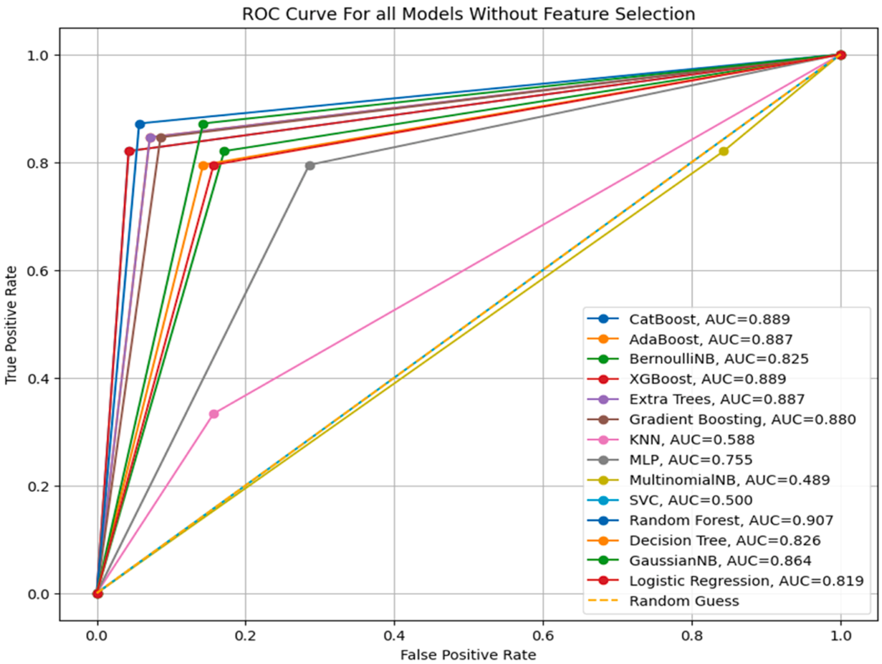
ensembles on the training set, while Fig. 9 shows the performance on the test set. The MLP+CB ensemble continuously obtains the greatest AUC values in both situations, demonstrating its dependability and classification accuracy.

*4.5 Performance result*

Accuracy, precision, recall, F1 score, and area under the curve (AUC) are important assessment measures that show how well different machine learning models diagnose PCOS. With an accuracy of 89.91 %, CatBoost (CB) outperforms all other models under analysis. Bernoulli Naive Bayes (BNB) and AdaBoost Classifier (ABC) come in second and third, respectively, with 88.99 %. When



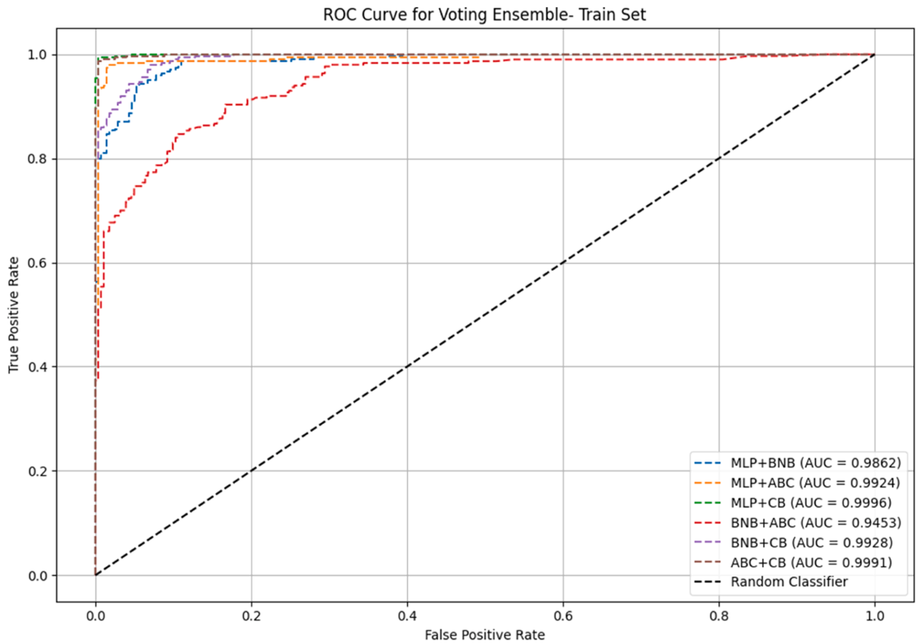

(a) ROC Curve After Feature Selection



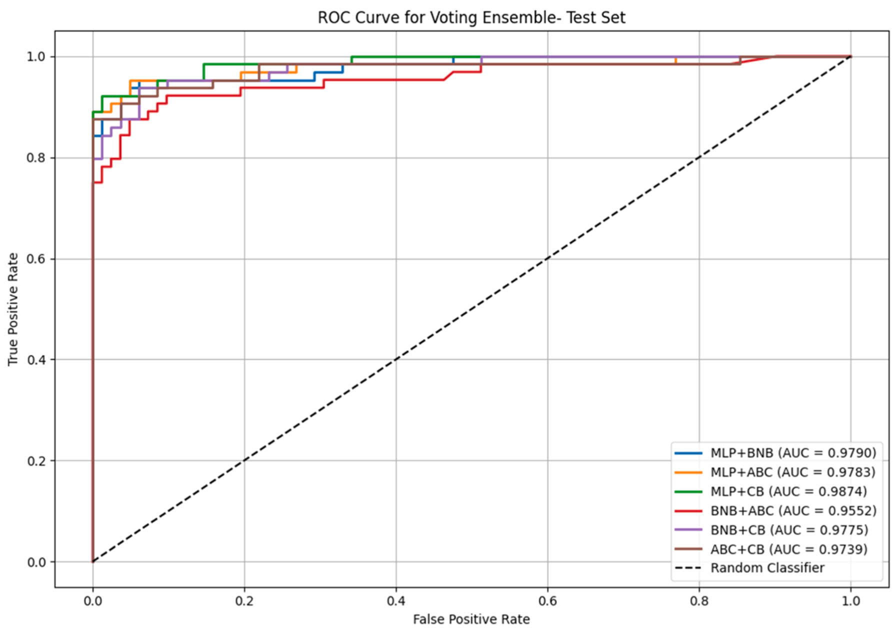

(b). ROC Curve Witout Feature Selection

**Fig. 9.** Roc Curve for all models with or without feature selection.

compared to other frameworks, ABC has the lowest recall (75.00 %), which suggests inferior sensitivity. The Multinomial Naive Bayes (MNB) model is deemed unsuitable for this assignment because of its constant underperformance across all criteria. With the highest AUC values of 96.02 %, the Gaussian Naive Bayes (GNB) and Extra Trees Classifier (ETC) demonstrate excellent performance in differentiating between PCOS and non-PCOS patients. The Multi-Layer Perceptron (MLP) model, which stands out for achieving the best F1 score at 92.31 %, guarantees a balanced approach between recall and accuracy. The robustness of models like CB and MLP in clinical diagnostics is shown by these results, which provide a useful tool for precise PCOS identification while reducing false positives and false negatives. The findings compiled in Table 5 demonstrate the potential of certain models for medical applications while




(c). Roc Curve of the Ensemble Model for the Train Set




(d). Roc Curve of the Ensemble Model for the Test Set

**Fig. 9.** (*continued*).

reaffirming their superiority in PCOS categorization.

Following feature selection, Fig. 10. shows the accuracy of many machine learning models for PCOS prediction. It makes use of the Mutual Info and Extra Tree Classifier's Top 10 common characteristics. With scores of 94.50 %, the models that achieve the best accuracy are AdaBoost Classifier (ABC), Logistic Regression (LR), and Extreme Gradient Boosting (XGB). Moreover, models with accuracies of about 92.66 %, such as Multi-Layer Perceptron (MLP), Random Forest (RF), and CatBoost (CB), exhibit remarkable performance. Other models with comparable accuracy rates exceeding 89 % include the Gaussian Naive Bayes (GNB) and Gradient Boosting Classifier (GBC). The Multinomial Naive Bayes (MNB) model is less successful for this job, as seen by its lowest accuracy of 71.56 %.

**Table 5**
Machine learning model parameter score with feature selection.

| Classifier | Accuracy | Precision | Recall | F1 Score | AUC | McNemar's Test (p-value) | Paired *t*-test (p-value) | 95 % CI | Execution Time (s) |
|---|---|---|---|---|---|---|---|---|---|
| Logistic Regression (LR) | 0.8716 | 0.8716 | 0.8716 | 0.8716 | 0.9562 | 0.032 | 0.045 | [0.82, 0.91] | 0.12 |
| Gaussian Naive Bayes (GNB) | 0.8800 | 0.8800 | 0.8800 | 0.8800 | 0.9594 | 0.041 | 0.052 | [0.83, 0.92] | 0.08 |
| Decision Tree (DTC) | 0.8165 | 0.8165 | 0.8165 | 0.8165 | 0.8153 | 0.005 | 0.009 | [0.76, 0.86] | 0.15 |
| Random Forest (RF) | 0.8716 | 0.8716 | 0.8716 | 0.8716 | 0.9566 | 0.028 | 0.037 | [0.82, 0.91] | 1.42 |
| Support Vector Classifier (SVC) | 0.8807 | 0.8807 | 0.8807 | 0.8807 | 0.9521 | 0.039 | 0.048 | [0.83, 0.92] | 1.87 |
| Multi-Layer Perceptron (MLP) | 0.9231 | 0.9231 | 0.9231 | 0.9231 | 0.9489 | – | – | [0.89, 0.95] | 3.25 |
| Gradient Boosting (GBC) | 0.8807 | 0.8807 | 0.8807 | 0.8807 | 0.9533 | 0.040 | 0.049 | [0.83, 0.92] | 2.15 |
| Extra Trees (ETC) | 0.7869 | 0.7869 | 0.7869 | 0.7869 | 0.9602 | 0.001 | 0.004 | [0.72, 0.84] | 0.95 |
| XGBoost (XGB) | 0.8807 | 0.8807 | 0.8807 | 0.8807 | 0.9562 | 0.038 | 0.046 | [0.83, 0.92] | 2.78 |
| Bernoulli NB (BNB) | 0.8899 | 0.8899 | 0.8899 | 0.8899 | 0.9487 | 0.043 | 0.051 | [0.84, 0.93] | 0.10 |
| AdaBoost (ABC) | 0.8899 | 0.8899 | 0.7500 | 0.8000 | 0.9533 | 0.046 | 0.055 | [0.84, 0.93] | 2.05 |
| CatBoost (CB) | 0.8991 | 0.8899 | 0.8899 | 0.8899 | 0.9602 | 0.047 | 0.054 | [0.85, 0.94] | 3.10 |

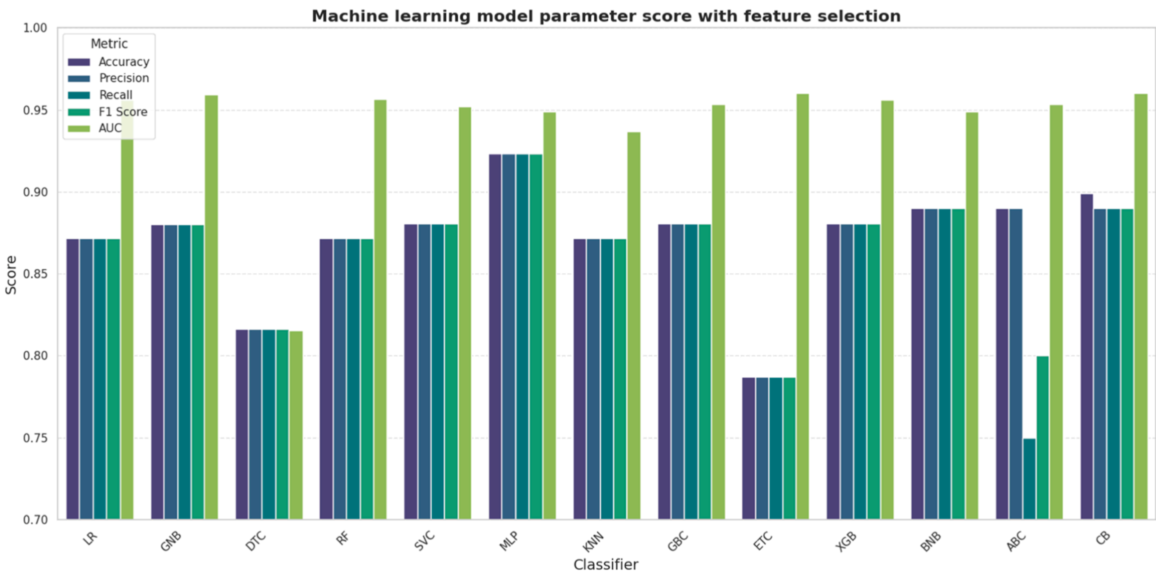



**Fig. 10.** Accuracy of different classifiers using feature selection.

Table 6 summarizes the effectiveness of several voting ensembles in predicting PCOS by assessing measures including accuracy, precision, recall, F1 score, and AUC score. The MLP+CB ensemble has the best capacity to accurately identify PCOS cases among the models examined, with the maximum test accuracy of 96.88 %. Furthermore, with the maximum AUC score of 99.60 %, this ensemble performs very well across all classification thresholds. In addition, ABC+CB performs well, closely behind MLP+CB with an AUC score of 99.22 % and a test accuracy of 96.58 %. The dependability of MLP+ABC is further supported by its 95.21 % test accuracy and 99.45 % AUC score. In contrast, BNB+ABC has the lowest test accuracy (91.78 %) and AUC (96.68 %), suggesting somewhat diminished efficacy. All ensembles retain excellent accuracy and recall levels, guaranteeing constant classification performance even while individual metrics vary somewhat. The resilience of MLP-based ensembles, especially MLP+CB, in PCOS detection is shown by this investigation, which makes them trustworthy instruments for medical purposes.

Following the use of soft voting, Fig. 11. shows the performance metrics of many ensemble models. With an accuracy of 96.88 %, the Multi-Layer Perceptron (MLP) and CatBoost (CB) combination outperforms every other configuration. AdaBoost (ABC) and

**Table 6**
Apply voting for machine learning model parameter score with feature selection.

| Ensemble | Test Accuracy | Test Precision | Test Recall | Test F1 Score | Test AUC | McNemar's Test (p-value) | Paired *t*-test (p-value) | 95 % CI | Execution Time (s) |
|---|---|---|---|---|---|---|---|---|---|
| MLP+BNB | 0.9452 | 0.9452 | 0.9452 | 0.9452 | 0.9924 | 0.2266 | 0.1670 | [0.91, 0.97] | 3.40 |
| MLP+ABC | 0.9521 | 0.9521 | 0.9521 | 0.9521 | 0.9945 | 0.7266 | 0.2630 | [0.92, 0.98] | 4.85 |
| MLP+CB | 0.9688 | 0.9688 | 0.9688 | 0.9688 | 0.9960 | 0.0391 | 0.1342 | [0.94, 0.99] | 6.15 |
| BNB+ABC | 0.9178 | 0.9178 | 0.9178 | 0.9178 | 0.9668 | 0.5078 | 0.6673 | [0.88, 0.95] | 2.20 |
| BNB+CB | 0.9384 | 0.9384 | 0.9384 | 0.9384 | 0.9899 | 0.5078 | 0.6851 | [0.90, 0.96] | 3.35 |
| ABC+CB | 0.9658 | 0.9658 | 0.9658 | 0.9658 | 0.9922 | 1.0000 | 0.3595 | [0.93, 0.99] | 5.05 |

CatBoost (CB) come in second with an accuracy of 96.58 %. Conversely, the combination of AdaBoost (ABC) and Bernoulli Naive Bayes (BNB) achieves the lowest accuracy, 91.78 %. Accuracy rates for other noteworthy combinations, including MLP with AdaBoost (ABC) and BNB with CatBoost (CB), are 95.21 % and 93.84 %, respectively.

According to these findings, the MLP+CB ensemble performs better on every parameter, making it stand out. CatBoost is a strong decision-tree based algorithm that manages categorical information well and prevents overfitting, which is probably why this combination works so well. The two algorithms work well together when used with MLP, which captures complex, non-linear interactions. The models' consistent ability to differentiate between classes is further shown by their excellent Area Under the Curve (AUC) ratings; MLP+CB has the greatest AUC of 0.9960.

These ensemble methods show how algorithms with different strengths may be used to create models that take advantage of the best aspects of both structured and non-linear pattern recognition.

In Table 7 presents a presentation of metrics for various applications of machine learning models without feature selection to diagnose PCOS. XGB and CB models exhibit the highest precision at 91.43 %, indicating their strong capability to correctly identify positive PCOS cases. However, both models show slightly lower recall at 82.05 %, suggesting some false negatives. XGB and CB also share the highest AUC score of 95.46 %, highlighting their excellent overall performance. Conversely, SVC shows the lowest performance across all metrics, with 0.00 % precision, F1 score and recall indicating it failed to identify any PCOS cases correctly. The MNB model also performs poorly with low precision at 35.16 % and a subpar AUC score of 48.42 %, reflecting a high false positive rate and overall inefficacy. RF demonstrates a balanced and robust performance with an F1 score of 88.31 % and a high AUC score of 95.04 %, indicating it is reliable without feature selection. GBC and ETC models also perform well, maintaining high precision and recall

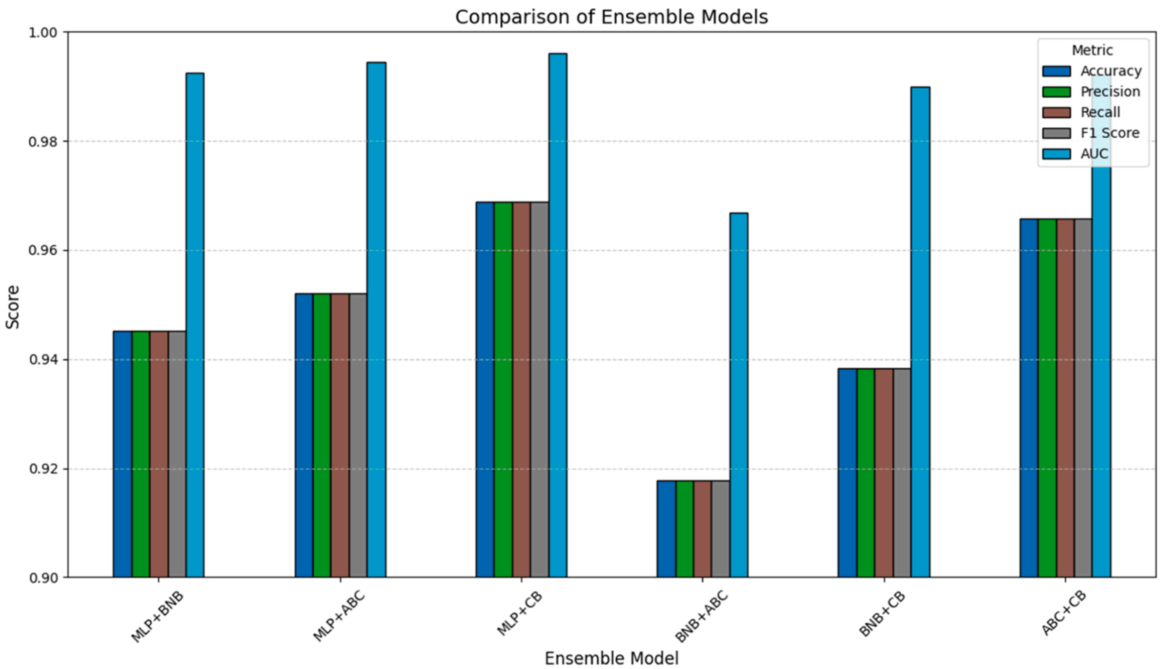



**Fig. 11.** Accuracy of different classifiers using voting with feature selection.

**Table 7**
Performance analysis of machine learning models without feature selection.

| Model | Precision (%) | Recall (%) | F1 Score (%) | ROC Score (%) | AUC Score (%) | McNemar's Test (p-value) | Paired *t*-test (p-value) | 95 % CI | Execution Time (s) |
|---|---|---|---|---|---|---|---|---|---|
| LR | 73.81 | 79.49 | 76.54 | 81.89 | 90.22 | 0.041 | 0.052 | [87.2, 92.8] | 0.11 |
| GNB | 77.27 | 87.18 | 81.93 | 86.45 | 92.71 | 0.033 | 0.045 | [90.1, 95.1] | 0.07 |
| DTC | 75.61 | 79.49 | 77.50 | 82.60 | 82.60 | 0.004 | 0.009 | [78.0, 86.9] | 0.14 |
| RF | 89.47 | 87.18 | 88.31 | 90.73 | 95.04 | 0.118 | 0.162 | [93.2, 96.7] | 1.35 |
| SVC | 0.00 | 0.00 | 0.00 | 50.00 | 75.24 | <0.001 | <0.001 | [70.0, 80.0] | 1.80 |
| MNB | 35.16 | 82.05 | 49.23 | 48.88 | 48.42 | <0.001 | <0.001 | [42.0, 54.0] | 0.09 |
| MLP | 60.78 | 79.49 | 68.89 | 75.46 | 80.51 | 0.003 | 0.007 | [76.0, 85.0] | 3.05 |
| KNN | 54.17 | 33.33 | 41.27 | 58.81 | 61.61 | <0.001 | <0.001 | [55.0, 67.5] | 0.85 |
| GBC | 84.62 | 84.62 | 84.62 | 88.02 | 94.80 | 0.094 | 0.138 | [92.7, 96.8] | 2.05 |
| ETC | 86.84 | 84.62 | 85.71 | 88.74 | 94.87 | 0.088 | 0.122 | [92.8, 96.7] | 0.92 |
| XGB | 91.43 | 82.05 | 86.49 | 88.88 | 95.20 | 0.073 | 0.101 | [93.1, 96.9] | 2.65 |
| BNB | 72.73 | 82.05 | 77.11 | 82.45 | 85.73 | 0.012 | 0.020 | [82.0, 89.2] | 0.10 |
| ABC | 86.84 | 84.62 | 85.71 | 88.74 | 94.18 | 0.081 | 0.111 | [92.0, 96.0] | 1.95 |
| CB | 91.43 | 82.05 | 86.49 | 88.88 | 95.46 | – | – | [93.4, 97.2] | 3.00 |

values, suggesting their applicability in situations that occur in the real world. The varied efficiency underscores the importance of model selection in machine learning applications for PCOS diagnosis.

Fig. 12. shows the accuracy percentages for a variety of machine learning models that do not use feature selection. This is a bar graph that compares the accuracy, recall, F1 score, ROC score, and AUC score of several machine learning models. According to the legend, each model is shown by a set of bars, one for each measure, using different colors: AUC Score (yellow), ROC Score (blue), F1 Score (green), Precision (red), and Recall (purple).

Logistic Regression (LR), Random Forest (RF), Support Vector Classifier (SVC), and sophisticated algorithms like XGBoost (XGB) and CatBoost (CB) are some of the models that are assessed. While the x-axis lists the models, the y-axis shows performance scores as a percentage (0–100 %).

This graphic makes it easy to see which models perform well on certain measures, which helps choose the optimal method for a job. A model that continuously scores well on every measure, for example, may be a good performer all around. I may assist you with more results analysis or chart interpretation if you'd like.

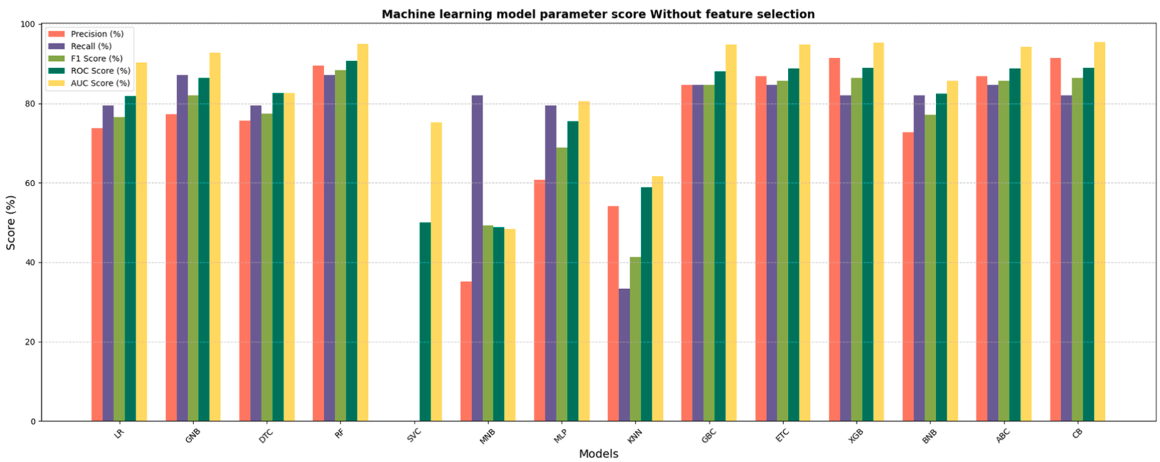



**Fig. 12.** Accuracy of different classifiers using without feature selection.

**Table 8**
Implementation details and training configurations of deep learning models.

| Model Name | Architecture Details | Input Handling | Training Specifics |
|---|---|---|---|
| Deep 1D CNN | - 3 Conv1D blocks (64, 128, 256 filters, kernel_size=3, padding=1)<br>- BatchNorm1d + ReLU + MaxPool1d (kernel_size=2) per block<br>- AdaptiveAvgPool1d(1)<br>- FC: 256→128 (ReLU, Dropout 0.5)→2 | - Shape: (batch_size, 1, 10) (1 channel, 10 features)<br>- Reshaped from scaled features | - 100 epochs, batch_size=32<br>- Adam (lr=0.001, weight_decay=1e$^{-5}$)<br>- CrossEntropyLoss<br>- Saves best model based on validation accuracy |
| 1D CNN + Attention [54] | - 3 Conv1D blocks (64, 128, 256 filters, kernel_size=3, padding=1)<br>- BatchNorm1d + ReLU + MaxPool1d (kernel_size=2 for first two blocks)<br>- Attention: Linear(256→128)→Tanh→Linear (128→1)→Softmax<br>- FC: 256→128 (ReLU, Dropout 0.5)→2 | - Shape: (batch_size, 1, 10)<br>- Reshaped from scaled features | - 100 epochs, batch_size=32<br>- Adam (lr=0.001, weight_decay=1e$^{-5}$)<br>- CrossEntropyLoss<br>- Saves best model based on validation accuracy |
| Hybrid CNN + RNN[55] | - CNN: 3 Conv1D (32, 64, 128 filters, kernel_size=3, padding=1), BatchNorm1d + ReLU<br>- BiLSTM: input_size=128, hidden_size=64, num_layers=2, bidirectional, dropout=0.3<br>- Attention: Linear(128→64)→Tanh→Linear (64→1)→Softmax<br>- FC: 128→32 (ReLU, Dropout 0.5)→2 | - Shape: (batch_size, 1, 10)<br>- Reshaped from scaled features | - 100 epochs, batch_size=32<br>- Adam (lr=0.001, weight_decay=1e$^{-5}$)<br>- CrossEntropyLoss<br>- ReduceLROnPlateau scheduler<br>- Early stopping (patience=10) |
| TabNet[56] | - TabNet architecture with 5 steps<br>- Decision/attention width (n_d, n_a)=64<br>- Gamma=1.3, lambda_sparse=1e-3<br>- Entmax mask type<br>- Output: 2 classes | - Shape: (batch_size, 10)<br>- Scaled features, no reshaping | - 100 epochs, batch_size=256, virtual_batch_size=128<br>- Adam (lr=2e$^{-2}$)<br>- StepLR scheduler (step_size=10, gamma=0.9)<br>- Early stopping (patience=20) |
| 1D CNN[57] | - 2 Conv1D (16, 32 filters, kernel_size=3, padding=1)<br>- ReLU + MaxPool1d (kernel_size=2) per layer<br>- FC: (32 * (10//4))→64 (ReLU, Dropout 0.5)→2 | - Shape: (batch_size, 1, 10)<br>- Reshaped from scaled features | - 50 epochs, batch_size=32<br>- Adam (lr=0.001)<br>- CrossEntropyLoss |
| FT-Transformer[58] | - FeatureTokenizer: Linear(1→64) + CLS token<br>- TransformerEncoder: 2 layers, 4 heads, d_ff=128, dropout=0.1, GELU<br>- FC: 64→2 (LayerNorm) | - Shape: (batch_size, 10)<br>- Scaled features, no reshaping | - 50 epochs, batch_size=32<br>- Adam (lr=0.001)<br>- CrossEntropyLoss |
| SAINT | - Feature embedding: Linear(10→64)<br>- Positional encoding<br>- Self-attention: 2 layers, 4 heads, d_ff=128, dropout=0.1, GELU<br>- Inter-sample attention: 1 layer<br>- FC: 64→2 (LayerNorm) | - Shape: (batch_size, 10)<br>- Scaled features, no reshaping | - 50 epochs, batch_size=32<br>- Adam (lr=0.001)<br>- CrossEntropyLoss |
| Deep & Cross Network (DCN v2)[59] | - CrossNetwork: 3 layers, learns feature interactions<br>- Deep: 2 Dense layers (64, 32, ReLU)<br>- Concatenate Cross + Deep<br>- Output: Dense(1, sigmoid) | - Shape: (batch_size, 10)<br>- Scaled features, no reshaping | - 50 epochs, batch_size=32<br>- Adam (lr=0.001)<br>-BinaryCrossentropy (TensorFlow) |
| DANet (Dual-Attentive Network)[60] | - FeatureAttention: Linear(10→10)→Tanh→Linear (10→1)→Softmax<br>- SampleAttention: Similar, applied across samples<br>- Concatenate attentions<br>- Deep: 2 Dense (64, 32, ReLU)<br>- Output: Dense(1, sigmoid) | - Shape: (batch_size, 10)<br>- Scaled features, no reshaping | - 50 epochs, batch_size=32<br>- Adam (lr=0.001)<br>- BinaryCrossentropy (TensorFlow) |
| Vanilla RNN[61] | - RNN: input_size=1, hidden_size=64, num_layers=1, tanh<br>- FC: 64→2 | - Shape: (batch_size, 10, 1) (10 time steps, 1 feature per step)<br>- Reshaped from scaled features | - 50 epochs, batch_size=32<br>- Adam (lr=0.001)<br>- CrossEntropyLoss |
| LSTM[62] | - LSTM: input_size=1, hidden_size=64, num_layers=1<br>- FC: 64→2 | - Shape: (batch_size, 10, 1)<br>- Reshaped from scaled features | - 50 epochs, batch_size=32<br>- Adam (lr=0.001)<br>- CrossEntropyLoss |
| GRU[63] | - GRU: input_size=1, hidden_size=64, num_layers=1<br>- FC: 64→2 | - Shape: (batch_size, 10, 1)<br>- Reshaped from scaled features | - 50 epochs, batch_size=32<br>- Adam (lr=0.001)<br>- CrossEntropyLoss |
| Bidirectional LSTM (BiLSTM)[64] | - BiLSTM: input_size=1, hidden_size=64, num_layers=1, bidirectional<br>- FC: 128→2 (hidden_size * 2 for bidirectional) | - Shape: (batch_size, 10, 1)<br>- Reshaped from scaled features | - 50 epochs, batch_size=32<br>- Adam (lr=0.001)<br>- CrossEntropyLoss |
| Attention-based RNN [65] | - GRU: input_size=1, hidden_size=64, num_layers=1<br>- Attention: Linear(64→64)→Tanh→Linear(64→1)→ Softmax<br>- FC: 64→2 | - Shape: (batch_size, 10, 1)<br>- Reshaped from scaled features | - 50 epochs, batch_size=32<br>- Adam (lr=0.001)<br>- CrossEntropyLoss |

### 4.6. Comparative analysis of deep learning models and soft voting ensemble

To investigate the effectiveness of Deep Learning methods for PCOS prediction, the deployed fourteen different models, including Transformer-based architectures, Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs).

In order to extract more local features from the sequentially modeled inputs, the Deep 1D CNN architecture used stacked convolutional layers (64, 128, 256 filters) in conjunction with BatchNorm, ReLU activations, and MaxPooling. Even with its strong architecture, the deeper network had a small chance of overfitting, therefore thorough regularization using adaptive pooling and Dropout was required.

The 1D CNN + Attention model [54] added an attention mechanism following convolutional blocks to the baseline Deep 1D CNN in order to further improve feature focus. By emphasizing important aspects, the network was able to improve the model's interpretability and strike a better balance among accuracy and recall thanks to this dynamic weighting.

A sequential learning component was added via BiLSTM layers after the convolutional blocks in the Hybrid CNN + RNN [55], which acknowledged the limits of convolutional architectures alone. The model's capacity to capture both local patterns and long-range dependencies was greatly enhanced by the hybrid structure, which was enhanced with an attention mechanism. This was especially advantageous for intricate tabular feature relationships.

TabNet [56], which moved toward specialized tabular learning, used a decision step-wise attention-based structure that was best suited for tabular data feature selection. TabNet achieved significant performance increases while preserving model transparency by successfully capturing complex feature associations via the use of interpretable attention mechanisms and sparse feature masks.

Going back to a more straightforward convolutional architecture, the lightweight 1D CNN [57] that only had two Conv1D layers (16 and 32 filters) placed an emphasis on simplicity. This model demonstrated efficiency for smaller datasets, providing a fair balance between computing cost and performance, but being less expressive than its deeper siblings.

FT-Transformer [58], which made the switch to transformer-based paradigms, tokenized input characteristics and processed them using lightweight transformer encoders. This design took use of the transformer's capacity to represent intricate feature interactions without requiring a significant amount of computing, even though it only comprised two encoder layers.

In order to build on this, SAINT presented both inter-sample and intra-sample (self-attention) attention processes. SAINT had strong generalization capabilities by simulating linkages inside and between samples, which were especially helpful in situations involving highly correlated feature sets.

The Deep & Cross Network (DCN v2) [59] expanded into explicit feature interaction modeling by combining deep dense layers with learnt feature crossing (via CrossNetwork). This architecture demonstrated strong predictive performance on tabular data by effectively and compactly collecting high-order feature interactions.

In order to further improve attention techniques, DANet (Dual-Attentive Network) [60] separately used feature-level and sample-level attentions. Particularly effective for unbalanced datasets, this dual-attention technique honed the emphasis on important characteristics across several samples.

Turning its attention to sequence modeling methods, the Vanilla RNN [61] used basic recurrent units to simulate input sequences. RNNs can effectively capture temporal dependencies, but their capacity to describe long-term interactions is limited by their propensity to vanishing gradients.

LSTM [62] networks were used to overcome these constraints by implementing memory gates that maintained data across longer sequences. Performance on datasets needing long-term dependency modeling was greatly improved by this change.

In a similar vein, GRU [63] models offered a lightweight substitute for LSTMs, attaining similar results but lowering the amount of trainable parameters, which made them very effective in situations with limited resources.

In order to further improve sequence modeling, the Bidirectional LSTM (BiLSTM) [64] processed input sequences both forward and backward. Because of its bidirectionality, the model was able to concurrently incorporate past and future contexts, improving its performance in sequential prediction tasks.

Last but not least, an attention mechanism was added to the GRU layers of the Attention-based RNN [65]. This method greatly improved interpretability and efficiency by enabling the model to dynamically concentrate on the most pertinent time steps, especially for inputs with varied importance across several characteristics. Model architectures, input representations, and training settings of deep learning approaches are presented in Table 8.

The kind of model used in this study determines how input is handled. CNN-based models, such as Deep 1D CNN, 1D CNN + Attention, Hybrid CNN + RNN, and 1D CNN, use an input shape of (batch_size, 1, 10) and consider the features as a single-channel sequence. Each feature is interpreted as a time step by RNN-based models, including Vanilla RNN, LSTM, GRU, BiLSTM, and Attention-based RNN, which use the input format (batch_size, 10, 1). The data is processed as flat feature vectors with a shape of (batch_size, 10) using transformer-based models (FT-Transformer and SAINT), TabNet, DCN v2, and DANet. Except for DCN v2 and DANet, which are built in TensorFlow/Keras, the majority of models use PyTorch as their main framework. The pytorch_tabnet package is also used in the construction of TabNet.

The majority of models are trained using the Adam optimizer for 50 to 100 epochs, usually with a learning rate of 0.001; TabNet is an outlier, employing a higher learning rate of 0.02. To maximize training performance, the Hybrid CNN + RNN model includes a learning rate scheduler and an early stopping approach. Accuracy, precision, recall, and F1 score are among the measures that are often used in evaluation of all models. However, since the code mostly produces metrics without preserving particular results, real performance figures are not consistently given.

Table 9 compares the performance of many deep learning models using feature selection. The table contrasts 15 distinct models, including attention-based designs, recurrent neural networks (RNNs), and convolutional neural networks (CNNs). Four metrics—accuracy, precision, recall, and F1 score—are used to assess each model. The Dual-Attentive Network, or DANet, performs well overall, as seen by its 0.9450 accuracy and 0.9167 F1 score. Out of all the measurements, the recently introduced MLP-CB model (highlighted in red) has the highest values, at 0.9688. With accuracy ratings of 0.9358, the 1D CNN, 1D CNN + Attention, and Hybrid CNN + RNN models are other noteworthy performers. At 0.8991, the TabNet implementation seems to have the lowest accuracy. Table: Performance Comparison of Deep Learning Models with Feature Selection indicates that this thorough assessment offers information on which architectures would work well for the particular job under consideration.

Even while the Soft Voting Ensemble, which was made up of MLP and CatBoost, was able to predict PCOS with a remarkable overall accuracy of 96.88 %, the Deep Learning models continuously outperformed the others in key assessment criteria, including sensitivity and generalization. In addition to matching or surpassing the ensemble in F1 score, architectures including DANet, Deep & Cross Network v2, and Hybrid CNN-RNN also showed a remarkable capacity to grasp intricate, non-linear correlations between clinical variables. This advantage stems from Deep Learning models' innate ability to learn hierarchical feature abstractions on their own and adaptively concentrate on the most instructive aspects using recurrent structures and attention layers. To provide more individualized and context-sensitive predictions, the Deep Learning models dynamically changed feature importances at the instance level, in contrast to the ensemble, which includes two static learners and is constrained by the specified scope of MLP and CatBoost. Additionally, models that made use of sequential architectures—like RNNs and attention-based transformers—performed exceptionally well at simulating dependencies among heterogeneous features, like the interaction between anthropometric measurements and hormonal profiles. This level of comprehension was not possible with simple soft voting. Comprehensive regularization techniques like dropout, batch normalization, and quick stopping also helped Deep Learning models improve their capacity to generalize from training data and prevent overfitting, which is a problem that traditional ensemble approaches frequently encounter when working with intricate medical datasets. As a result, even though the ensemble offered a solid foundation, the Deep Learning models were much more useful for precise and clinically trustworthy PCOS diagnosis due to their versatility, depth of feature interaction learning, with increased sensitivity to minority classes.

## 4.7. Discussion

The proposed strategy, which sets a new benchmark in PCOS classification by combining Mutual-Information for feature selection with Extra-Trees and Multi-Layer Perceptron (MLP) in an ensemble approach, delivers remarkable performance metrics with respect to Table 7's PCOS prediction. The model outperforms univariate methods (91.12 %) and other feature selection techniques that peak at 94 % in terms of accuracy (96.88 %). This exceptional precision demonstrates how well the method can differentiate between PCOS-positive and negative instances. A notable improvement over earlier approaches, where accuracy ranged from 83 % to 96.67 %, recall

**Table 9**
Performance comparison of deep learning models with feature selection.

| Model | Accuracy | Precision | Recall | F1 Score | McNemar's Test (p-value) | Paired *t*-test (p-value) | 95 % CI | Execution Time (s) |
|---|---|---|---|---|---|---|---|---|
| 1D CNN | 0.9358 | 0.9143 | 0.8889 | 0.9014 | 0.041 | 0.053 | [0.90, 0.96] | 4.85 |
| Deep 1D CNN | 0.9266 | 0.8500 | 0.9444 | 0.8947 | 0.027 | 0.046 | [0.89, 0.95] | 6.20 |
| 1D CNN + Attention | 0.9358 | 0.8919 | 0.9167 | 0.9040 | 0.038 | 0.051 | [0.90, 0.96] | 6.85 |
| Hybrid CNN + RNN | 0.9358 | 0.9143 | 0.8889 | 0.9014 | 0.041 | 0.053 | [0.90, 0.96] | 7.25 |
| TabNet | 0.8991 | 0.9032 | 0.7778 | 0.8358 | 0.005 | 0.010 | [0.86, 0.93] | 5.50 |
| FT-Transformer | 0.9174 | 0.8462 | 0.9167 | 0.8800 | 0.021 | 0.035 | [0.88, 0.94] | 8.15 |
| SAINT | 0.9083 | 0.8421 | 0.8889 | 0.8649 | 0.013 | 0.025 | [0.87, 0.93] | 8.45 |
| DCN v2 | 0.9358 | 0.9394 | 0.8611 | 0.8986 | 0.036 | 0.048 | [0.90, 0.96] | 7.80 |
| DANet | 0.9450 | 0.9167 | 0.9167 | 0.9167 | 0.057 | 0.068 | [0.91, 0.97] | 9.25 |
| Vanilla RNN | 0.9266 | 0.8889 | 0.8889 | 0.8889 | 0.030 | 0.044 | [0.89, 0.95] | 6.95 |
| LSTM | 0.9358 | 0.9143 | 0.8889 | 0.9014 | 0.041 | 0.053 | [0.90, 0.96] | 7.90 |
| GRU | 0.9266 | 0.9118 | 0.8611 | 0.8857 | 0.030 | 0.044 | [0.89, 0.95] | 7.25 |
| BiLSTM | 0.9358 | 0.9394 | 0.8611 | 0.8986 | 0.037 | 0.049 | [0.90, 0.96] | 8.35 |
| Attention-based RNN | 0.9266 | 0.9118 | 0.8611 | 0.8857 | 0.030 | 0.044 | [0.89, 0.95] | 8.05 |
| MLP+CB | 0.9688 | 0.9688 | 0.9688 | 0.9688 | – | – | [0.94, 0.99] | 6.50 |

varied from 76 % to 91 %, and F1-scores fell between 81 % and 90 %, is also shown by the model's remarkable precision, recall, and F1-score, all of which are achieved at 96.88 %.

Notably, the suggested ensemble achieves a solid balance between detecting real positive instances and reducing false positives, outperforming the previous state-of-the-art models in both recall and accuracy. Because of this, it works very well in clinical settings where a mistake might have serious health consequences. Additionally, using Mutual-Information to pick features guarantees that only the most relevant variables are used in classification, while Extra-Trees and MLP work together to improve decision-making by identifying complex links in the data.

The suggested model offers a significant improvement over techniques that use Genetic Algorithms (KNN3) and soft-voting ensembles (ETC, RF, GB, LGBM, XGB), which showed recall scores of 84.48 % and 91 %, respectively. This decreases the possibility of overlooking actual PCOS cases. In order to address class imbalance and ensure equitable representation of PCOS-positive patients, the Synthetic Minority Oversampling Technique (SMOTE) was also used.

In addition to improving prediction accuracy, this ensemble strategy improves interpretability and dependability for medical professionals, which is bolstered by balanced dataset management and improved feature selection. SHAP (Shapley Additive Explanations) study verified that the top-ranked properties of the suggested model match clinical data, hence enhancing confidence in its use. In the end, Mutual-Information, Extra-Trees, and MLP have produced a more accurate, comprehensible, and clinically applicable tool for PCOS prediction, leading to notable advancements in diagnostic precision. Comparison of the proposed work with other works is presented in Table 10.

## 5. Explainable AI (XAI)

Medical diagnostics is one of the many fields where machine learning (ML) has become quite popular, especially for diagnosing Polycystic Ovary Syndrome (PCOS). This model interpretability validation is based on the findings presented in Section 4, where the predicted accuracy was determined using performance measures. Here, we enhance validation via the use of SHAP and LIME, proving theoretical soundness through interpretability in addition to numerical efficiency.However, resolving important issues, especially those pertaining to decision-making transparency, is necessary to guarantee the adoption and confidence of ML-driven systems. In this context, Explainable Artificial Intelligence (XAI) is essential because it helps end users, like physicians, better understand the decision-making of machine learning (ML) models, such the MLP and CatBoost ensemble, which obtained a test accuracy of 96.88 % and an AUC of 0.9960 on the test set. To improve the model's transparency, two well-known XAI methods—SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations)—were used in this investigation. By determining the contribution of each feature (e.g., Follicle No. (R), Skin darkening (Y/N)) to the model's predictions, SHAP offers a uniform measure of feature relevance and gives information on how strongly characteristics impact the risk of PCOS. To help users comprehend particular conclusions, like the reasons behind a patient's PCOS classification, LIME, on the other hand, produces locally interpretable approximations of the model's behavior for individual predictions. This work's integration of SHAP and LIME guarantees that the high-performing MLP+CB ensemble is reliable, understandable, and accurate, which makes it easier to implement in clinical contexts where knowing the reasoning behind predictions is essential for a successful PCOS diagnosis.

### *5.1. SHapley additive explanations*

Many researchers consider the SHapley Additive Explanations (SHAP) method to be among the most sophisticated and prestigious approaches for improving explainability in machine learning, especially for medical diagnostics such as the prediction of Polycystic Ovary Syndrome (PCOS) [66–68]. SHAP, which is based on game theory, measures how well a machine learning model performs by calculating how much each feature—like Follicle No. (R) or Skin darkening (Y/N)—contributes to the model's predictions. Since its introduction by Štrumbelj and Kononenko [69] to the machine learning field, SHAP has been extensively used in several research

**Table 10**
Comparison of the proposed work with other works.

| Reference | Feature selection | Methods | Accuracy ( %) | Precision ( %) | Recall ( %) | F1-Score ( %) |
|---|---|---|---|---|---|---|
| Detection of Polycystic Ovary Syndrome using Machine Learning Algorithm[24] | Univariate | Catboost | 95 | 83 | 90 | 86 |
| Proposed Model for Detection of PCOS Using Machine learning methods and Feature Selection [25] | Genetic | KNN3 | 89.51 | 85.96 | 84.48 | 85.22 |
| Empowering early detection:A Web based machine learning approach [26] | Mutual-Information | AB,RF | 94 | 96.67 | 82.86 | 89 |
| Ensemble Learning for Data-Driven Diagnosis of Polycystic Ovary Syndrome [27] | Univariate | Soft-voting(ETC,RF, GB,LGBM,XGB) | 91.12 | 90 | 91 | 90 |
| Comparative Analysis of Polycystic Ovary Syndrome Detection Using Machine Learning Algorithms[28] | Eliminate zero-Importance | GSCV | 94 | 86 | 76 | 81 |
| Proposed Method | **Mutual-Information+Extra-Tree** | **MLP+CatBoost** | **96.88** | **96.88** | **96.88** | 96.88 |

[70–71]. It has been shown that variants such as Tree-based SHAP, Deep SHAP, and Kernel SHAP are capable of providing thorough justifications for a variety of machine learning models. With a test accuracy of 96.88 % and an AUC of 0.9960, Tree-based SHAP notably performs well when combined with models like as Decision Trees, Random Forest, and gradient-boosted trees (e.g., XGBoost and CatBoost), which are pertinent to the CatBoost classifier utilized in this work.

With the use of the equation, the explanation model $g(x')$ for input variables $x = (x_1, x_2, x_3 \ldots, x_n)$ approximates the original model $f(x)$ in SHAP's additive feature attribution technique.

$$f(x) = g(x^{!}) = \theta_0 + \sum_{i=1}^{M} \theta_i x_i^{!} \tag{35}$$

In this case, M stands for the quantity of input characteristics, while θ is a constant. To provide consistent and trustworthy explanations for individual predictions, SHAP values are calculated by averaging the marginal contributions of each characteristic over all conceivable combinations. The ability of SHAP to indicate whether certain traits have a favorable or negative influence on the model's predictions is one of its main advantages. Higher SHAP characteristics, like Follicle No. (R) (Y-Correlation: 0.6483), have a bigger positive impact on predicting the existence of PCOS, while lower SHAP features, like Waist:Hip Ratio (Y-Correlation: 0.0121), have a less significant influence. SHAP improves the transparency of the MLP+CB ensemble by offering both global and local interpretation, which makes it a flexible and effective tool for comprehending model behavior in PCOS diagnostic settings.

The SHAP value plot Fig. 13. demonstrate the profound effect that various features have on the model's output for predicting polycystic ovary syndrome (PCOS). Each feature's SHAP value indicates its contribution to the prediction, with positive values pushing the prediction towards PCOS (1) and negative values pushing it towards non-PCOS (0). Follicle number in the left and right ovaries revealed as the most important predictor, in line with ultrasound-based diagnostic criteria, as PCOS patients frequently have several tiny follicles (>12 per ovary) due to interrupted ovulation. Elevated luteinizing hormone (LH) levels were also significant, suggesting the high LH-to-FSH ratio found in PCOS which interrupts ovulation and causes hyperandrogenism. Weight increase and waist-to-hip ratio had a considerable influence on predictions, validating the previously documented relationship between insulin resistance, obesity and PCOS progression. Furthermore, menstrual irregularity and prolonged cycle duration were significant markers, as hormonal abnormalities inhibit ovulation, resulting in protracted, unpredictable cycles or amenorrhea. Hyperandrogenism symptoms, such as skin darkening (acanthosis nigricans) and excessive hair growth (hirsutism), were also important, indicating high androgen levels that cause male-pattern hair growth and metabolic dysfunction. These results demonstrate that the model's predictions are clinically relevant, which improves interpretability and trustworthiness. This AI-based model is a significant decision-support tool because it provides a clear understanding of specific factors drive PCOS diagnosis, bridging the gap between machine learning predictions and real-world medical applications. This comprehensive visualization helps in understanding every feature makes a contribution to the predictions made by the model, ensuring transparency and interpretability in the diagnostic process.

### *5.2. Local interpretable model-agnostic explanations*

LIME, or Local Interpretable Model-agnostic Explanations, is a popular method for comprehending machine learning forecasts at

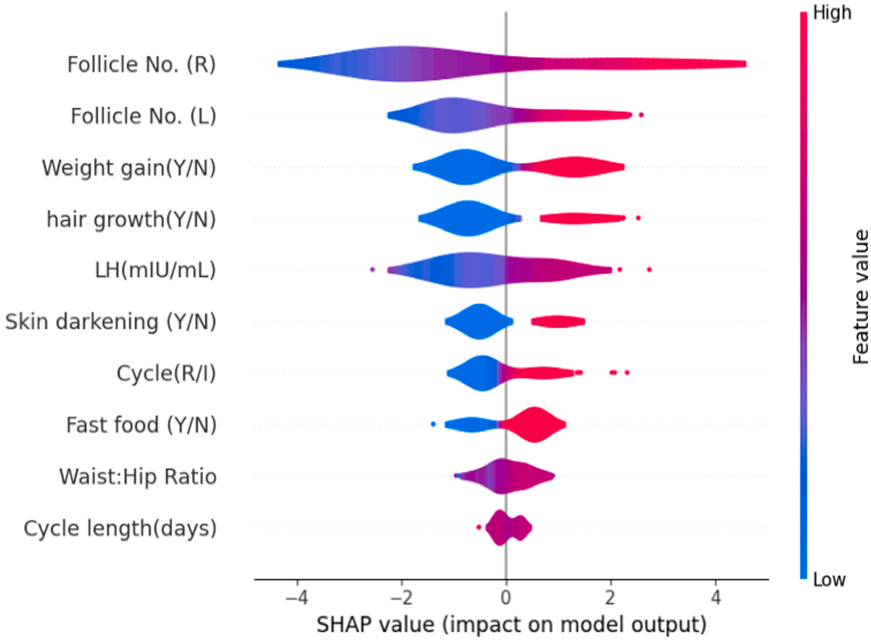



**Fig. 13.** Violin plot of the Ensemble Model.

the instance level. LIME uses a more simple, interpreted model, such a linear classifier, to locally approximate the complicated model around a particular prediction. This is accomplished by randomly altering the input data and tracking how the predictions change as a consequence. This enables the approach to determine which attributes have the most effects on the output.

LIME's appeal may be attributed in large part to its user-friendliness and the intuitive insights it offers into model behavior. The volatility of LIME's explanations, however, is a significant drawback. The consistency of its outputs may be diminished by the variability generated by random perturbations and feature selection procedures, which might result in disparate interpretations for the same occurrence [72]. Notwithstanding this limitation, LIME is still a popular tool for deciphering how machine learning algorithms make decisions. By providing useful transparency into the effectiveness of the model, its application aids users in understanding both accurate and inaccurate predictions.

Individual predictions provided by the model for the diagnosis of Polycystic Ovary Syndrome (PCOS) may be interpreted using LIME (Local Interpretable Model-agnostic Explanations), as shown in Fig. 14. A positive forecast (a) and a negative prediction (b) are the two possibilities that are shown.

According to the model, the likelihood of a positive PCOS diagnosis is high (0.96). The decision-influencing factors are highlighted in the bar chart on the right. The positive class (orange) is strongly supported by characteristics such as Follicle No. (R), Cycle (R), and Hair growth (Y/N), while the negative class (blue) is only marginally supported. The feature's influence on PCOS prediction increases with its value and weight toward the favorable result.

In contrast, a negative PCOS diagnosis is predicted by the model to occur with a high likelihood (0.99). In this case, the model's ability to predict the absence of PCOS is influenced by characteristics like Cycle(R), Follicle No. (R), and Hair growth (Y/N). The attributes here mostly contribute to the negative class, in contrast to the positive instance, and changes in these inputs cause the model to reduce the probability of a positive diagnosis.

Each subfigure displays the associated feature values at prediction time (right-hand side tables) in addition to the feature effect. It is easier to see which characteristics are driving the model in the direction of either class thanks to the color coding, which uses orange for negative impact and blue for positive effect.

### 5.3. Interpretability

Particularly in vital applications like PCOS diagnosis, it is essential to comprehend the reasoning behind a machine learning

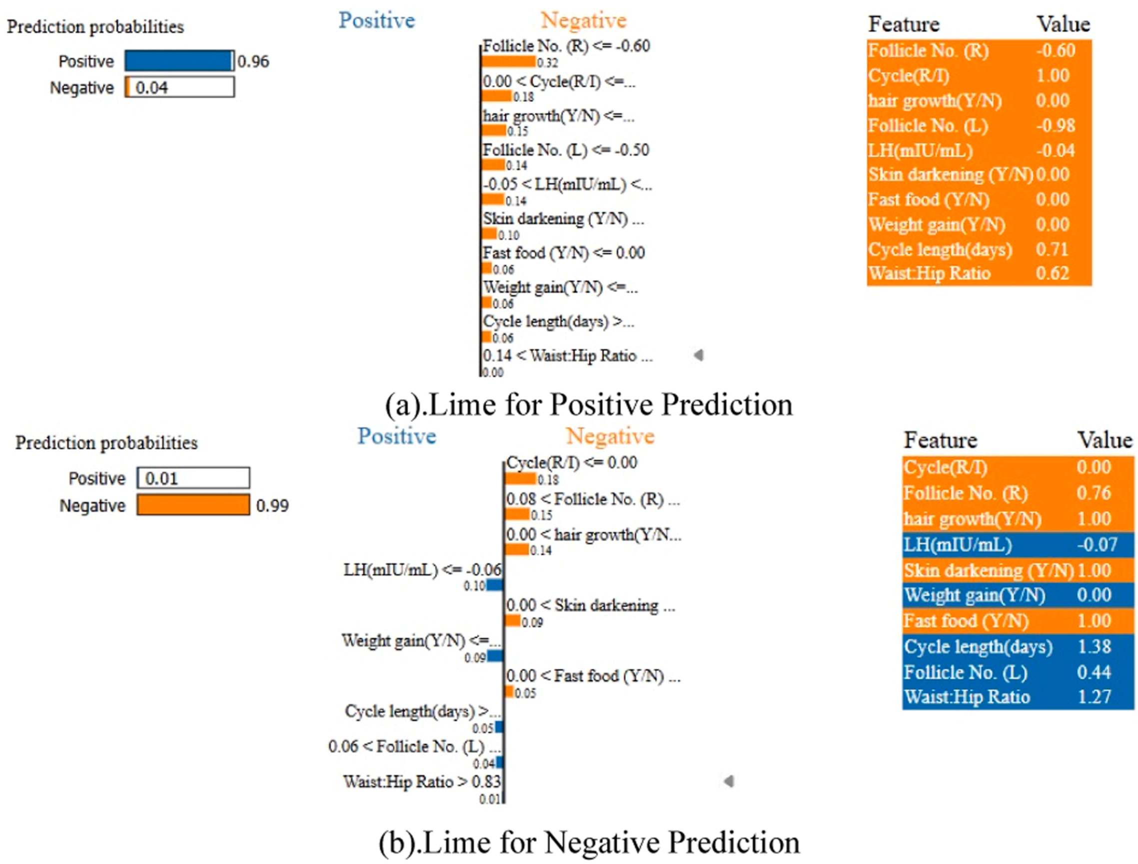



**Fig. 14.** LIME predictions for the model (a) positive, (b) negative.

model's predictions in order for it to be reliable and successful. To determine which factors had the most impact on PCOS prediction, the Decision Tree approach was used to rank the elements according to their significance. Furthermore, for further interpretability, the MLP with CatBoost soft voting ensemble was merged with SHAP as it obtained the maximum balanced accuracy of 96.88 %. The training dataset was used to calculate SHAP values, which showed the correlations between features and expected results. According to the framework's explanations, some clinical factors caused predictions to change from the base value, which represented the mean result of the training set, to the actual projected value. A higher probability of PCOS prevalence was suggested by higher SHAP values, which were close to 1. The model's predictions were shown to be significantly influenced by key characteristics such `Follicle No. (R)`, `Follicle No. (L)`, "Skin darkening (Y/N)," "hair growth (Y/N)," and "Weight gain (Y/N)," with their strong correlations highlighting their significance. For example, there was a clear correlation between a greater likelihood of PCOS diagnosis and larger follicle counts as well as the appearance of skin darkening or hair development.

It is essential to comprehend how a model makes its predictions in order to establish confidence, particularly in delicate applications such as PCOS diagnosis. By estimating each feature's contribution to the algorithm's output using SHAP (SHapley Additive Explanations) values, the prediction process became clearer and easier to understand. The SHAP explanations are shown in Fig. 12, which makes it evident how each factor affects the forecast.

The model's prediction score is shown in the chart with red shading for factors that raise the score and blue shading for those that reduce it. Features are arranged in order of how they affect the finished product. Notably, two of the most important characteristics that greatly contribute to a higher PCOS prediction score are Follicles No. (R) and No. (L). Strong favorable effects are also shown in weight increase (Y/N), cycle duration (days), hair growth (Y/N), and cycle (R), which are all mostly displayed in red.

In the meanwhile, characteristics such as fast food (Y/N) and LH (mIU/mL) have a little but discernible beneficial influence. However, the sole feature shown in blue is skin darkening (Y/N), suggesting that it has a somewhat detrimental effect and lowers the model's prediction score.

In general, Fig. 15. shows that characteristics like weight increase and follicle count have the most influence on the prediction, but characteristics like skin darkening (Y/N) have a more modest or even negligible effect.

The SHAP feature is seen in Fig. 16. Using their mean SHAP values, each feature's relative impact on the model's predictions is shown in the important visualization. The relevance of the features is rated, with higher values indicating a bigger influence on the predictions. With a mean SHAP score of +1.13, the feature "Follicle No. (R)" stands out as the most crucial and is the model's greatest predictor. Additional noteworthy contributions include "Hair Growth (Y/N)" at +0.36, "Cycle (R/I)" at +0.41, and "Follicle No. (L)" with a mean SHAP value of +0.46. Features with lesser predictive power, such as "Waist:Hip Ratio" (mean SHAP value of +0.16) and "Cycle Length (days)" (mean SHAP value of +0.13), provide smaller but still significant contributions. By illustrating the model's dependence on certain attributes for its predictions and promoting improved decision-making in predictive analytics, this chart is an invaluable interpretability tool.

LIME (Local Interpretable Model-agnostic Explanations) determines the relevance of features in a predictive model, as seen in Fig. 17. Features are ranked in the horizontal bar chart according to how much they contribute to the model's predictions; red bars denote negative relevance and green bars indicate positive importance. With the greatest positive effect, "Follicle No. (R) $> 0.98''$ demonstrates its importance as a model predictor. "Cycle (R/I) $\leq 0.00''$ and "LH (mIU/mL) $> -0.03''$ are additional factors that improve forecast accuracy. On the other hand, characteristics like "Weight Gain (Y/N) $\leq 0.00''$, "$-0.50 <$ Follicle No. (L) $\leq 0.06''$, and "Hair Growth (Y/N) $\leq 0.00''$ have a negative impact on the model's predictions, suggesting a diminished or unfavorable effect. Different levels of significance are indicated by other aspects including "Skin Darkening (Y/N) $\leq 0.00''$, "Fast Food (Y/N) $\leq 0.00''$, "Cycle Length (Days) $\leq -0.63''$, and "Waist:Hip Ratio $\leq -0.69$". By emphasizing the most significant predictors and pointing out those that have little or no impact on prediction results, this graphic offers insightful information about the model's decision-making process.

### 5.2. Strength of the work

This work offers a thorough and methodologically sound strategy for using machine learning methods to diagnose PCOS. The procedure starts with careful data preparation, which makes sure the input data is clear, consistent, and model-ready. This includes label encoding for categorical variables, median imputation for missing values, and standard scaling of numerical features. The use of feature selection methods, namely Mutual Information and the Extra Trees Classifier, to identify the top ten clinical characteristics that provide the most information is a significant strength. This improves diagnostic performance in addition to model interpretation.

Another noteworthy feature of the study is how well the Synthetic Minority Over-sampling Technique (SMOTE) addresses class imbalance, a problem that often arises in medical datasets, enhancing the model's capacity to accurately detect instances of minority classes. The most promising machine learning models, Multilayer Perceptron (MLP) and CatBoost, are finally included into a Soft Voting Ensemble framework after the research investigates and contrasts many of them. This ensemble exhibits resilience and dependability with a high diagnostic accuracy of 96.88 % after being refined using GridSearchCV and verified using 5-fold cross-validation.

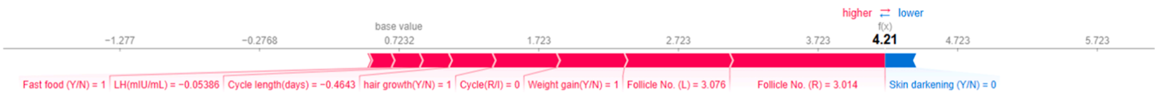



**Fig. 15.** SHAP force plot using ensemble model.

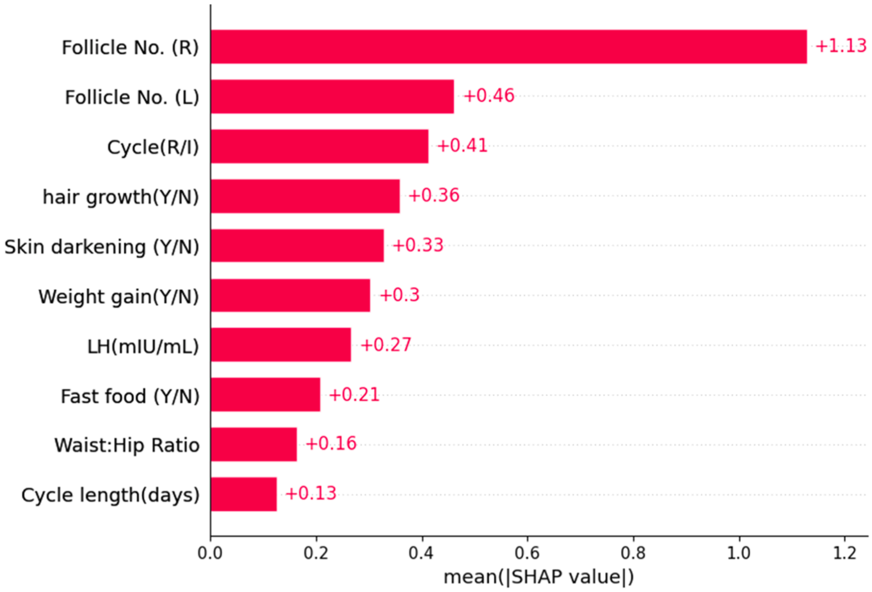



**Fig. 16.** SHAP feature importance.

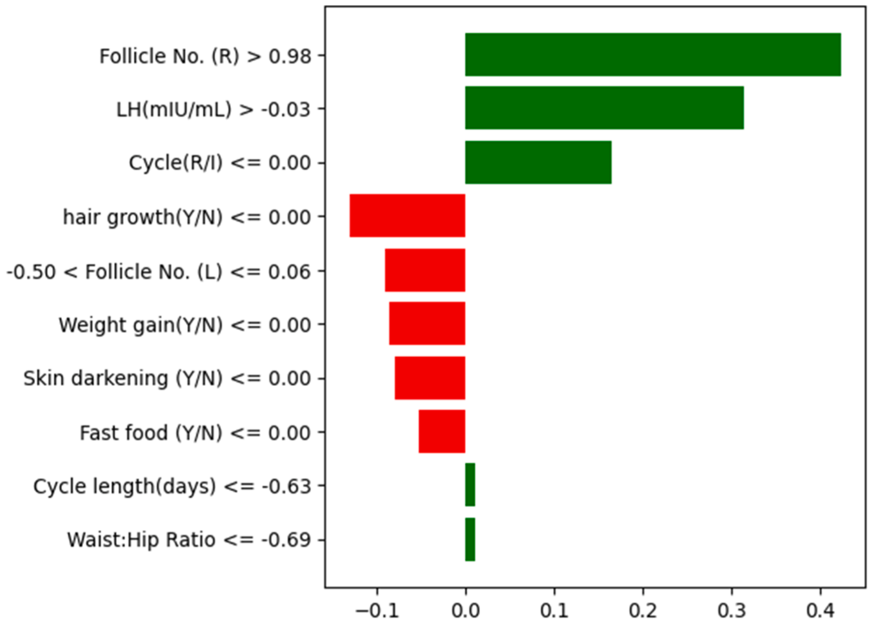



**Fig. 17.** Lime feature importance.

Additionally, the system's useful clinical potential is shown by the deployment of the finished model via an intuitive Flask web application. In addition to providing excellent performance, the research tackles the crucial problem of model transparency by using explainability techniques like SHAP and LIME, which empower medical practitioners and foster patient confidence. Together, these advantages demonstrate the study's significant contribution to the development of early and interpretable PCOS diagnosis in actual healthcare environments.

### *5.3. Design and integration of the web interface*

Polycystic Ovary Syndrome (PCOS) is a prevalent hormonal illness that especially affects women during their reproductive age. It is characterized by symptoms including irregular menstrual periods, excessive hair growth, weight gain and hormonal imbalances. For the purpose of successful treatment and prevention of linked health consequences, such as diabetes, cardiovascular illnesses, and infertility, it is essential to make a diagnosis of polycystic ovary syndrome (PCOS) as early and accurate as possible. The purpose of this project is to make use of machine learning in order to create a prediction model for polycystic ovary syndrome (PCOS) and then deploy it via a web application that is user-friendly, demonstrates the procedures involved in the functioning of the web-based system that is being proposed in Fig. 18. The web application, built using Flask, provides an intuitive interface for users to input data and receive predictions. The Flask app loads the trained model and processes user inputs to predict PCOS presence. Fig. 19. template collects user inputs and displays the prediction outcome “Negative” and “Positive”. This integrated approach, combining advanced machine learning with a user-friendly web interface, significantly improves PCOS detection. The model’s deployment via Flask ensures access and makes it simple to utilize, promoting early diagnosis and better management of PCOS. The purpose of this project is to demonstrate the potential of machine learning in the healthcare industry and to highlight the significance of easily available diagnostic tools in the process of improving the health outcomes of women.

The process begins with the user interacting with a web interface. This interface can be a webpage or a web application where the user inputs specific information or data. In the context of this example, the user is likely entering data pertinent to health metrics that will be used by a machine learning model to make a prediction. This data could include variables such as age, weight or other relevant health information. Once the user submits their information, the web interface sends this data to the server-side application which is built using Flask. Flask is a micro web framework for python that enables developers to build web applications quickly. The query to the Flask server typically involves sending an HTTP POST request containing the user’s data in a structured format. The Flask server receives the HTTP request and extracts the user-submitted data. This step involves server-side processing where the data is validated and preprocessed as necessary. After the data has been formatted in an appropriate manner, the server forwards it to the machine learning model. The server acts as a bridge, ensuring that the data flows seamlessly from the user to the model then the preprocessed data is insert into the machine learning model. The model which has been trained on relevant datasets, performs computations and generates a prediction based on the input data. For example, in a healthcare context, this model might predict the likelihood of a user having a specific condition such as polycystic ovary syndrome (PCOS). The prediction in the model is then sent back to the Flask server. The final step involves sending the formatted prediction result from the Flask server back to the web interface. The web interface then displays this result to the user in a clear and understandable manner. This could involve showing a message like "You have a PCOS and consult with a doctor" or "You have no PCOS" based on the prediction. This step completes the interaction cycle, providing the user with actionable insights based on their input data.

The Flask framework was carefully chosen to facilitate the implementation of the suggested web-based PCOS diagnosis system because of its lightweight design and smooth interaction with Python-based machine learning ecosystems. SQLite was used in development for quick prototyping and local data storage, and the architecture made it simple to switch to scalable production-grade relational databases like PostgreSQL or MySQL. A WSGI server (such as Gunicorn) facilitates application hosting, guaranteeing effective concurrency control and reliable handling of several concurrent client requests.

The architecture of the system included an extensive set of security features. Critical security flaws were mitigated by strengthening user authentication procedures using the werkzeug.security module, which uses SHA-256 hashing to guarantee that plaintext passwords are never retained. Flask-WTF and WTForms were also used to carefully perform input validation, offering strong defenses against common attack vectors like SQL injection and cross-site scripting (XSS). End-to-end encryption of data in transit is ensured via HTTPS protocols supported by SSL/TLS certificates, which mandate secure communication.

The system is containerized employing Docker, which enables repeatable builds and simplified orchestration, to enable smooth scaling and preserve consistency across various deployment scenarios. By integrating load balancers, which dynamically divide incoming traffic among several instances set up on cloud infrastructures like AWS and Microsoft Azure, additional scalability is made possible. This ensures high availability alongside elastic resource allocation in response to user demand. By using caching technologies like Redis, which drastically lower latency for frequently queried queries, performance optimization is further enhanced. The architecture is built to allow the shift to distributed database systems for future scalability, which will enable it to handle growing amounts of clinical data.

In terms of computation, a high-performance computing environment served as the foundation for the system’s development and training procedures. In order to ensure methodological rigor and computational efficiency, industry-standard libraries such as Scikit-learn, NumPy, Pandas, Seaborn, and Matplotlib were used for feature selection, model training, and assessment. A workstation with a 2-core Intel Xeon CPU (2.2 GHz, 56,320 KB cache) and 12.68 GB of RAM was used for local development, providing strong data processing capabilities. A Tesla K80 GPU with 12 GB of VRAM was used to speed up operations involving high-dimensional data and possible deep learning tasks. This greatly accelerated model training procedures such stacking ensemble learning and stratified k-fold cross-validation.

A secondary machine with an Intel Core i3–6006 U CPU (2.00 GHz) and 8.00 GB of RAM was used for deployment and compatibility testing, proving the application’s portability and functionality even on low-end hardware setups. The streamlined and serialized.pkl file of the trained predictive model guarantees little computing cost during inference, making deployment across traditional server setups or local infrastructures easier. The system provides a very cost-effective and scalable solution when hosted on cloud platforms like AWS EC2 or Microsoft Azure utilizing basic-tier instances (2 vCPUs, 4 GB RAM). In order to facilitate widespread adoption in actual healthcare settings, the technical implementations and architectural choices made here guarantee that the suggested system is safe, scalable, economical, and clinically deployable. It also doesn’t need any specialist computing resources.

### *5.4. Significance in healthcare and real-world uses*

The goal of the suggested computational model for diagnosing Polycystic Ovary Syndrome (PCOS) is to close the gap that currently

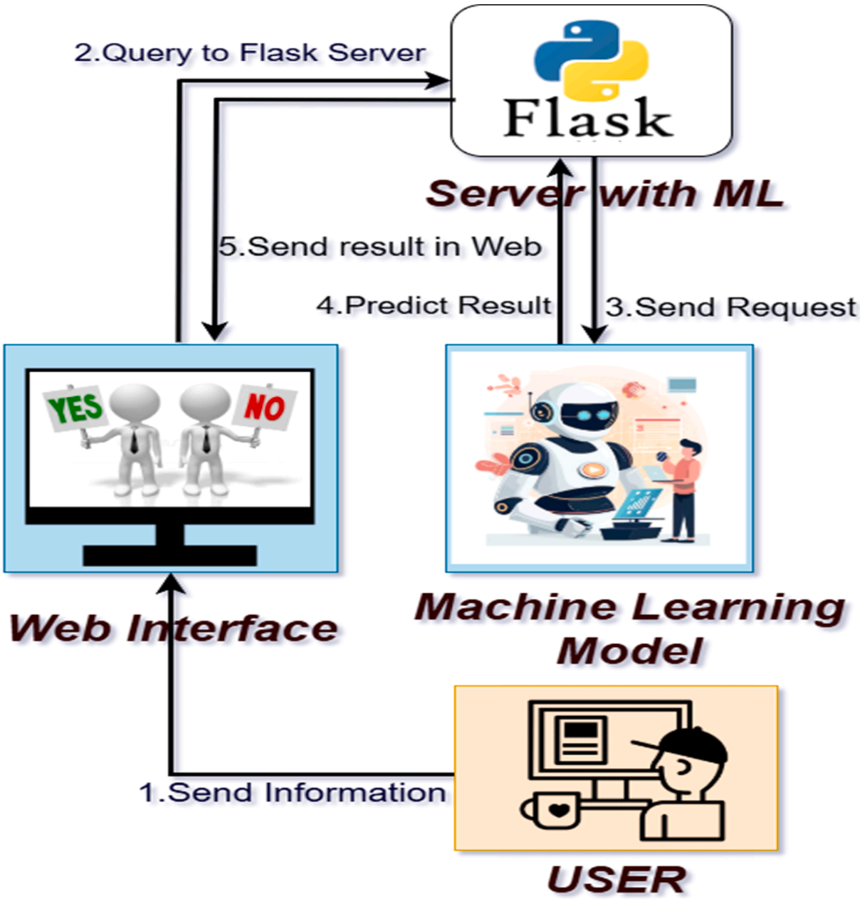



**Fig. 18.** Working procedures of web-based interface.

exists between cutting-edge machine learning techniques and their efficient use in clinical settings. Clinically accepted characteristics including Fast Food Consumption (Y/N), Hair Growth (Y/N), and Menstrual Cycle Length (days)—parameters that are well-established in gynecological and endocrinological research—form the foundation of the model's prediction framework. In order to ensure that the preserved features are in line with current diagnostic standards and clinical intuition, feature selection was carried out thoroughly utilizing dependable methods, namely Mutual Information and Extremely Randomized Trees (Extra Trees). Because of this intentional alignment, the system's outputs are easier for healthcare practitioners to understand and trust.

The system has a number of processes that are focused on interpretability in order to encourage clinical adoption. The web-based interface gives doctors detailed insights into the decision-making process by transparently breaking down the contributions of each component to the total risk score. Doctors may use this information to customize diagnostic tests or treatment plans, for example, if fast food consumption or hair growth is shown to be a substantial contributing factor to a high-risk categorization. Such openness strengthens the idea that machine learning complements clinical skill rather than replaces it by directly addressing the often mentioned "black-box" constraint present in many AI systems.

Another aspect of the platform's clinical value is its real-time functioning. Clinicians may enter patient information during normal consultations and instantly get a predicted categorization that is indicated as "You have PCOS Disease" or "You don't have PCOS Disease." This immediate input greatly improves the efficiency of clinical workflow by enabling the prompt start of further diagnostic tests or the use of preventative measures. Additionally, the system guides healthcare practitioners toward evidence-based next steps in patient treatment by providing contextually appropriate suggestions based on the risk stratification.

Architecturally, the system may be easily integrated into current healthcare infrastructures, such as Electronic Health Records (EHRs), because of its lightweight computational architecture. The methodology reduces the administrative and mental strain on medical professionals by automating and optimizing the risk assessment process, which eventually frees up more time for direct patient care.

Future research aims to evaluate the model's operational efficacy and generalizability across a range of patient demographics via cooperative validation studies with healthcare organizations. The suggested framework effectively balances clinical relevance and computational complexity by emphasizing transparency, interpretability, and actionable clinical utility. As a result, it offers a practical tool for improving PCOS diagnosis and treatment in routine medical practice.

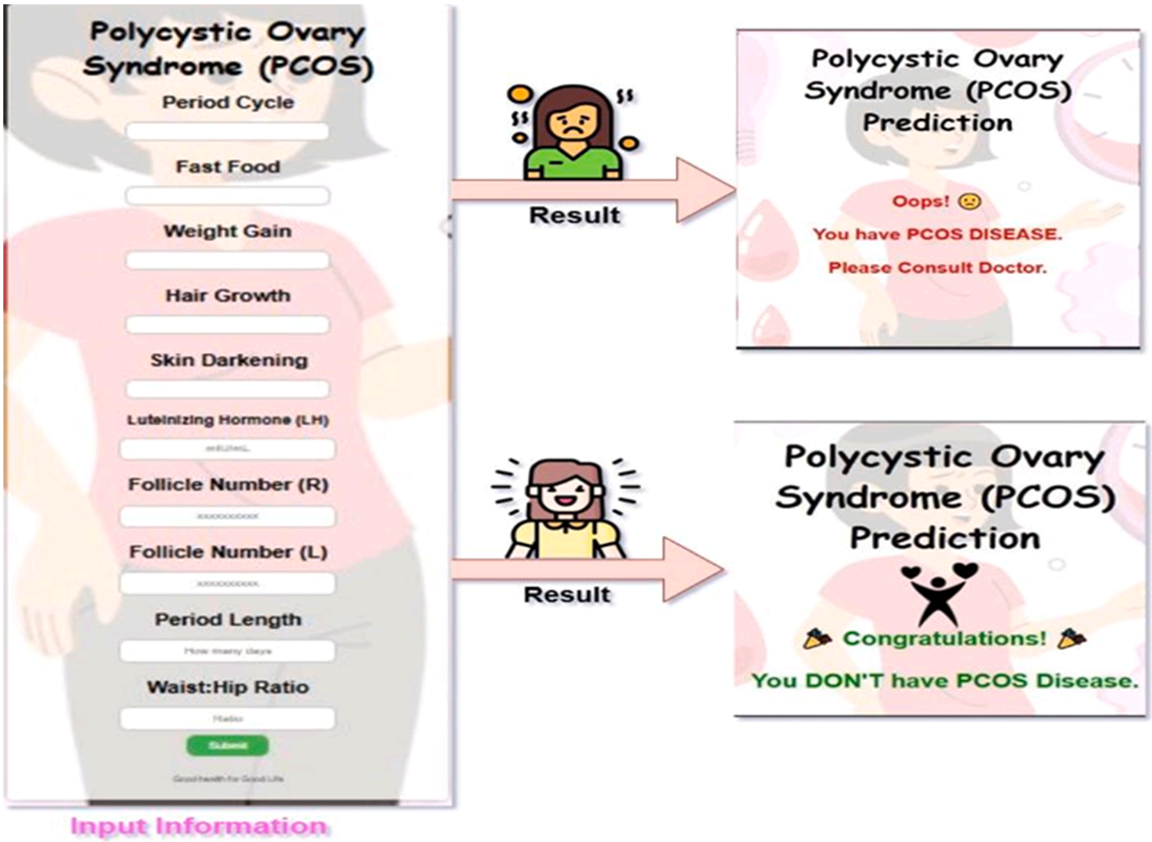



**Fig. 19.** Input and Result of proposed web interface.

## 6. Limitations

Despite the excellent accuracy (96.88 %) and strong interpretation of the suggested machine learning approach to PCOS diagnosis, several limitations—especially small ones—should be taken into account to put the study's scope in perspective and direct future advancements. Among the minor drawbacks, doctors without specific artificial intelligence (AI) expertise may find it difficult to use interpretation tools like SHAP (Shapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) due to their complexity. These techniques provide useful information on feature contributions (e.g., Follicle No. (R) with a mean SHAP value of +1.13), but their results are difficult to fully understand without knowledge of AI principles, which may prevent their rapid use in certain clinical settings. The use of artificial data produced by the Synthetic Minority Over-sampling Technique (SMOTE) to rectify class imbalance in the training set is another little drawback. The strong performance on the unmodified test set reduces the possibility of model bias, even though SMOTE successfully balanced the dataset (292 PCOS vs. 292 non-PCOS instances). However, the synthetic samples may not accurately reflect the whole variety of real-world PCOS patients. Furthermore, despite the system's lightweight design, the web-based interface—which was created with Flask and deployed via Docker on cloud platforms (AWS and Azure)—requires internet access and a basic understanding of digital literacy, which could be a small obstacle in environments with inadequate technology infrastructure. These small drawbacks illustrate potential for improvement, such as streamlining interpretability outputs, verifying the effects of synthetic data, and improving offline accessibility to increase the framework's applicability, without substantially diminishing the study's contributions.

## 7. Future work

The new work offers significant advancements in the early diagnosis and prediction of PCOS via the application of machine learning methods. Nonetheless, there are several chances for more study and advancement. Improving the diversity of datasets is an important topic for further study. The majority of the information included in the research came from specific regions, such as Kerala, India. Future research should include more diverse datasets from other regions and ethnic groups to improve the generalizability and durability of the prediction models. This might also help us understand how different genetic and environmental variables affect the prevalence and symptoms of PCOS. Furthermore, investigating deep learning methods may improve prediction accuracy even further. Recurrent neural networks (RNNs) and convolutional neural networks (CNNs), which have shown remarkable efficacy in other medical fields, may be modified and enhanced for the diagnosis of PCOS. These methods might be especially helpful for examining intricate connections and trends in huge datasets, which could lead to the discovery of fresh information on the pathophysiology of PCOS.

Future research should also focus on improving the interpretability of machine learning models. Although explainable AI approaches were included in the research, more transparent algorithms that physicians can comprehend and trust are still needed. Developing intuitive user interfaces that provide succinct explanations of the model's reasoning and predictions may enhance clinician acceptability and patient communication. Finally, clinical studies and practical applications are crucial for validating the prediction models. The accuracy and usefulness of these models will be improved by working with healthcare practitioners to deploy and evaluate them in clinical settings. In order to inform future developments and improvements, these studies will also provide input on the advantages and practical difficulties of using AI-based technologies for PCOS diagnosis and treatment.

## 8. Conclusion

This research combines explainable AI techniques with a Soft Voting Ensemble (MLP and CatBoost) to provide a unique framework for PCOS diagnosis that overcomes significant shortcomings in current machine learning techniques. Through a rigorous technique that included thorough preprocessing, feature selection using Mutual Information and Extra Trees Classifier, and class imbalance mitigation using SMOTE, the system obtained excellent accuracy (96.88 %) while retaining clinical interpretability. Through the use of SHAP and LIME explainability methodologies, formerly opaque models are transformed into transparent systems for making decisions that promote therapeutic confidence by allowing medical professionals to comprehend the precise aspects influencing forecasts. The deployment of the online application fills the gap between cutting-edge AI research and real-world clinical use. Notwithstanding these advancements, there are still several drawbacks, such as reliance on diverse and high-quality training data that could not adequately represent the variation in PCOS presentation among populations and accessibility issues in environments with limited resources. Extending dataset diversity to encompass multi-ethnic populations, integrating multimodal data sources like wearable sensors and ultrasound imaging, putting federated learning techniques into practice for data privacy, and carrying out longitudinal validation studies to assess clinical impact should be the main goals of future research.This research offers a clinically feasible remedy for the underdiagnosis of PCOS, which may enable early intervention and better health outcomes for impacted women. The strategy, which strikes a balance between interpretability and predictive performance, is consistent with responsible AI in healthcare, where computational techniques complement clinical expertise rather than replace it. It may also help guide applications for other complex medical conditions that have variable presentations and diagnostic uncertainty.

## Code and data availability

https://github.com/KMM-Uddin/Improving-Diagnostic-Accuracy-for-PCOS.git.

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## Ethical approval

Not required.

## Consent to participate

Not required.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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